Non-Stationarity in Financial Time Series: A Unifying Survey on Drift Detection, Adaptive Learning, and Evaluation

Davi M. Cabral[a,*], Adriano M. A. Lima[a], Gustavo H. F. M. Oliveira[b], Adriano L. I. Oliveira[a]

[a] *Centro de Informática (CIn), Universidade Federal de Pernambuco, Recife, PE, Brasil*
[b] *Sistemas de Informação, Universidade Federal de Alagoas, Penedo, AL, Brasil*

Abstract

Predictive and decision models in finance are typically validated under assumptions of distributional stability over the evaluation window. In deployment, those assumptions fail: the data-generating process undergoes structural change—breaks, regime transitions, and drift—that can invalidate conditional relationships, degrade calibration, and amplify tail risk precisely when decisions are most consequential. Despite a large literature, results remain hard to reconcile across econometrics, statistical monitoring, and machine learning due to divergent terminology and incompatible evaluation protocols. This survey aims to overcome the fragmentation and provides three concrete tools for research and deployment under non-stationarity in financial time series: **(1)** a unified taxonomy of drift and regime change, **(2)** a pipeline that connects representation, detection, and adaptation choices, and **(3)** an evaluation playbook that supports apples-to-apples comparison. We align terms such as structural breaks, regimes, concept drift, and dataset shift, and propose a taxonomy along temporal, statistical, spatial, and ontological axes to describe real drift scenarios consistently. Using this lens, we review drift-aware representations, change detection methods, and continuous adaptation strategies—from classical sequential monitoring and segmentation to Bayesian, multivariate, and embedding-based out-of-distribution approaches. We then consolidate evaluation guidance spanning detection delay, false-alarm control, computational cost, and finance-specific utility. Finally, we highlight emerging directions (foundation models, multimodal context, parameter-efficient adaptation) and open challenges in benchmark design and reliable online calibration.

Keywords:
financial time series, non-stationarity, concept drift, change-point detection, adaptive learning, evaluation protocols

1. Introduction

Financial markets are dynamic and inherently non-stationary systems, in which the data generation process alternates between distinct statistical regimes due to macroeconomic shocks, liquidity crises, regulatory changes, and shifts in agent behavior [**? ? ? ?**]. This phenomenon appears in the literature under various terms—including concept drift, regime change, structural breaks, and heteroscedasticity—and directly affects core financial tasks [**? ? ? ? ?**] such as price forecasting, risk management, order execution, and portfolio allocation [**? ?**], creating persistent challenges for predictive modeling in real-world, high-stakes deployments.

In practice, non-stationarity progressively degrades the predictive performance and calibration of forecasting models [**? ? ? ?**]. Under prequential evaluation, this degradation manifests as systematic increases in forecast error, unstable decision thresholds, and delayed reactions to new regimes—often precisely when errors are most costly in terms of risk exposure and economic decision-making [**? ? ?**]. These effects expose the limitations of stationarity-based assumptions and motivate approaches capable of anticipating and responding to distribution shifts as they occur.

Addressing these challenges requires coupling drift-aware representations with timely change detection and continuous adaptation [**? ? ? ? ?**], while evaluation must explicitly account not only for predictive accuracy but also for detection delay and computational

*Corresponding author. Email: dmc6@cin.ufpe.br
Email addresses: dmc6@cin.ufpe.br (Davi M. Cabral), amal@cin.ufpe.br (Adriano M. A. Lima), gustavo.oliveira@penedo.ufal.br (Gustavo H. F. M. Oliveira), alio@cin.ufpe.br (Adriano L. I. Oliveira)

constraints [? ?]. In practice, the effectiveness of this coupling depends critically on how data are represented, since informative representations are necessary to make distribution shifts observable rather than masking them [? ?]. This dependency naturally shifts attention to the role and design of financial data representations under non-stationarity.

In non-stationary financial environments, data representations must capture both endogenous market dynamics and exogenous drivers. Embeddings learned from price and volume series summarize endogenous behavior, enabling comparisons across historical periods and the identification of latent regimes beyond classical indicators [? ?]. However, market behavior is also shaped by external forces—such as macroeconomic announcements, geopolitical events, and firm-level disclosures—which are not observable in price data alone. Incorporating exogenous information through multimodal inputs (e.g., news, textual reports, and economic indicators) adds economic context to detected shifts [? ? ?], yielding representations that are more expressive under concept drift and better suited to distinguishing transient fluctuations from structurally meaningful regime transitions.

With such enriched representations, continuous adaptation becomes feasible. Model behavior can be updated through classical mechanisms—including forgetting factors and regime-switching structures [? ? ? ? ?]—as well as through modern architectures that support lightweight domain adaptation, such as specialized financial foundation models [? ?]. In both cases, contextualized representations enable models to respond more selectively and interpretably to evolving conditions, balancing stability and responsiveness to improve robustness in online forecasting and decision-making. However, the practical relevance of these adaptive gains is inseparable from how model performance is defined and measured over time.

Evaluation therefore emerges as a third essential dimension, complementing representation and adaptation. To ensure practical effectiveness, evaluation protocols must jointly account for predictive performance, detection speed, false-alarm control, computational budgets, and economic impact, reinforcing the need for realistic and comparable benchmarks [? ? ? ? ? ?]. Despite advances along these axes, progress remains uneven and often confined to specific methodological traditions, limiting comparability and cumulative evidence.

At a broader level, the theoretical landscape remains fragmented [? ? ? ?], with modeling assumptions, methodological choices, and evaluation practices diverging across communities. As a result, machine learning, econometrics, and quantitative finance have largely evolved in parallel, with limited cross-fertilization and incompatible frameworks. The absence of a shared taxonomy and standardized evaluation hinders cumulative evidence and slows the adoption of drift-aware methods in practice, particularly in real-time and high-stakes financial settings [? ?].

With the aim of addressing these gaps, we formulate the following research questions: (RQ1) How does the literature define and categorize the different forms of non-stationarity in financial time series? (RQ2) How can financial series be represented by integrating endogenous and exogenous information to support the detection and interpretation of regime changes? (RQ3) How can distribution shifts be automatically detected over time? (RQ4) How can model learning be adapted to distribution shifts continuously and effectively? (RQ5) How can detection and adaptation systems be evaluated under non-stationarity using appropriate metrics and protocols? (RQ6) How can detection and adaptation methods be benchmarked under non-stationarity? (RQ7) What are the current limitations and future research directions?

In addressing these questions, this work makes four contributions. (i) We introduce a unified taxonomy of drift and regime change phenomena in financial time series, aligning terminology across machine learning, econometrics, and quantitative finance (RQ1, Section 2). (ii) We structure the literature around a five-pillar pipeline—non-stationarity characterization, drift-aware representations, change detection, continuous adaptation, and evaluation—and review representative approaches for representation learning, change detection, and adaptation (RQ2–RQ4, Sections 3, 4, and 5). (iii) We summarize evaluation metrics, experimental protocols, and benchmarking practices used in financial time series (RQ5–RQ6, Sections 6 and 7). (iv) Finally, we synthesize current limitations and outline open research directions (RQ7, Sections 8 and 9).

1.1. Research methodology and related work

This survey was developed through an iterative strategy for literature retrieval, reading, and synthesis, guided by keyword searches and by the temporal chaining of contributions. Initial searches were carried out in broad-coverage digital libraries and academic aggregators (e.g., ScienceDirect, IEEE Xplore, ACM Digital Library, SpringerLink, and Google Scholar), with a primary focus on publications from 2000 to 2025. Whenever needed to contextualize modern formulations of *drift* and regime changes, we also included a small set of earlier foundational references (e.g., in sequential analysis, econometrics, and change-point detection).

Search strings combined three intersecting fronts: (i) non-stationarity terminology (e.g., *concept drift*, *distribution shift*, *regime change/shift*, *structural break*, *change point*); (ii) operational mechanisms (e.g., *drift detection*, *change-point detection*, *sequential tests*, *adaptive/continual learning*, *test-time adaptation*, *retraining*, *ensembles*); and (iii) financial contexts and realistic protocols (e.g., *algorithmic trading*, *risk management*, *portfolio*, *volatility*, *market microstructure*, *backtesting*, *transaction costs*). Beyond surveys and reviews, we also considered experimental and methodological studies whenever they contributed directly to answering the RQs and to structuring the survey's pipeline (foundations, detection, adaptation, and evaluation).

The corpus was progressively refined based on topical relevance and usefulness for the proposed synthesis, complemented by *snowballing* (backward and forward) from seminal works, recent surveys, and recurring references. This process resulted in a superset of 289 unique candidate references (deduplicated across sources and rounds).

A central element of this refinement was examining the future-work sections of the retrieved papers and, whenever possible, checking whether the proposed directions were later investigated or addressed. This "from-future-to-present" tracking served as a structured scan of gaps and adjacent research lines: by relating explicit recommendations to subsequent evidence, we were able to make persistent gaps explicit and map emerging themes correlated with regime changes and *concept drift* (e.g., GNNs, foundation models, multimodality, *deep reinforcement learning*, and knowledge representation). After an initial relevance screening and consolidation to remove near-duplicates and out-of-scope items, this process yielded a shortlist of 220 references.

For scope reasons and to preserve coherence with the RQs and the proposed conceptual pipeline, part of these adjacent themes was deliberately deprioritized and not discussed in depth (e.g., topics centered on MLOps/operational monitoring, temporal *fairness*, privacy/federated learning, and *vintage data*/real-time data revisions, among others), remaining as opportunities for future work. The final manuscript therefore cites 174 references.

Finally, the selected studies were organized according to their predominant role in an end-to-end pipeline for financial systems under non-stationarity: (1) foundations and terminology harmonization; (2) representation and context modeling; (3) detection of distributional changes and regimes; (4) adaptation mechanisms; and (5) evaluation and benchmarking. We emphasize that the goal is not to exhaustively cover all publications in this rapidly growing area, but to highlight seminal and representative contributions that clarify concepts, design trade-offs, and practical implications for financial deployment.

2. Drifts and Regimes Foundations and Taxonomy

This section seeks to answer the question: *how does the literature define and categorize the different forms of non-stationarity in financial time series?* Different fields — statistics, econometrics, machine learning, finance, and the natural sciences — propose complementary taxonomies to address non-stationarity. In this context, there is a broad consensus that non-stationarity refers to variation, over time, in the statistical or structural properties of a data-generating process [**? ? ? ? ?**]. However, beyond this high-level definition, these fields differ in the way such changes in the data distribution are formalized, categorized, and analyzed.

Motivated by these differences in conceptualization and emphasis, we present an ontology, illustrated in Figure 1, that describes the non-stationarity problem in financial time series [**? ? ? ? ?**] along four main classification axes: (i) a temporal axis, which characterizes when and how changes unfold over time; (ii) a statistical axis, which specifies which properties of the data-generating process are affected; (iii) a spatial axis, which describes where changes manifest within the data structure; and (iv) an ontological axis, which distinguishes the nature and formal status of the change. Finally, we discuss the causal axis, which identifies the underlying drivers of non-stationarity, including exogenous shocks, endogenous feedback mechanisms, and adversarial or institutional effects, linking observed drifts and regime transitions to their sources.

2.1. Temporal axis

In the specific case of time series, non-stationarity may affect the mean and variance of the very relationship between variables or the mechanism that produces them. One way to characterize these effects is by focusing on how the changes unfold in relation to the timing and temporal shape of the change in the process, that is, how and when the change occurs, for example, as illustrated in Figure 2. Common types include abrupt drift, gradual drift, incremental (or continuous) drift, and recurrent (or seasonal) drift [**? ? ? ? ? ?**], but some surveys also highlight *blips* (transient deviations / outliers) to differentiate short-lived noise from structural

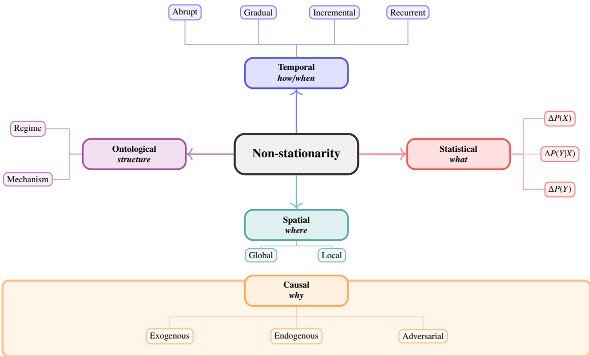

Figure 1: Extended taxonomy of non-stationarity along five axes: temporal (how/when), statistical (what changes), spatial (where), ontological (structure/state), with causal drivers (why) as the foundational layer.

changes [? ? ? ?], which we define below to clarify their temporal dynamics.

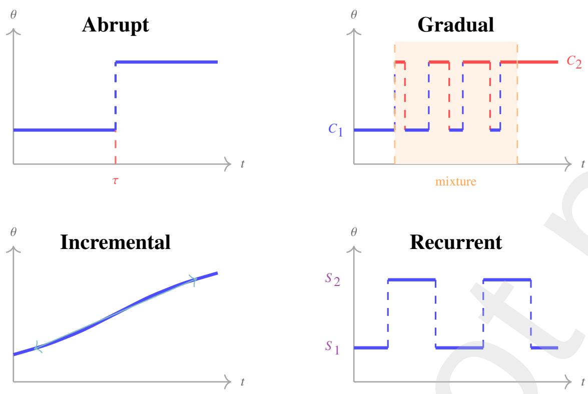

Figure 2: Temporal morphology of drifts: **Abrupt**—sudden change at changepoint τ; **Gradual**—transition interval where old and new concepts coexist and switch with each other gradually; **Incremental**—continuous parameter drift without stable plateaus; **Recurrent**—alternation between previously observed states.

- **Abrupt drift**: sudden change that establishes a new level almost instantaneously. For instance, the revelation of a major accounting fraud at a large corporation may trigger an immediate market-wide sell-off, abruptly shifting volatility regimes, risk premia, and correlation structures across related sectors, with the new price dynamics and investor sentiment establishing themselves within hours or a single trading session.

- **Gradual drift**: transition interval in which observations from the old and new concepts coexist, with a fuzzy boundary (a mixture of states for some period). A canonical example is the slow transition from an upward-trending regime to a downward-trending regime, as market conditions progressively weaken (from a bull market—a period of broadly rising prices and optimistic sentiment—to a bear market—a period of broadly declining prices and pessimistic sentiment[1]).

- **Incremental / continuous drift**: continuous drift of parameters without a clear breakpoint or stable plateaus, i.e., without discrete states. Examples include long-run structural (secular) trends, such as the multi-decade decline in long-term interest rates [? ?](Fig. 3A), driven by slow-moving forces in saving and investment rather than a single abrupt shock.

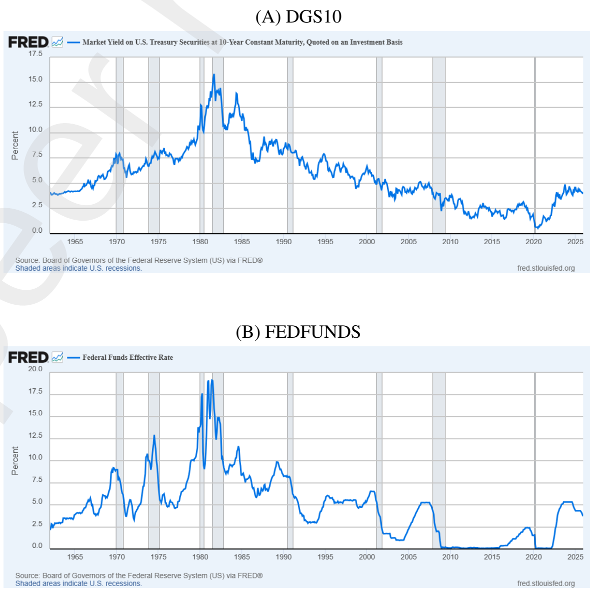

Figure 3: U.S. interest-rate series at macro *monthly* time scales (sample: 1962-01 to 2025-12). (A) 10-year Treasury constant-maturity yield (DGS10; end-of-period at this frequency). (B) Effective federal funds rate (FEDFUNDS), summarizing the stance of U.S. monetary policy. Over this period, the 10-year yield declines from around 14% in the early 1980s to about 8% in the early 1990s, around 5–6% in the late 1990s, near 2% in the mid-2010s, and below 1% in 2020. Policy-rate movements in (B) shape short-term financing conditions and can transmit to longer maturities in (A) through expectations and term premia, although the 10-year yield also reflects inflation expectations and broader risk compensation. Source: FRED [? ?].

- **Recurrent / seasonal drifts**: unlike gradual drift, which describes a one-way transition where old

[1] In equity markets, a common rule of thumb defines a bull (bear) market as a rise (fall) of about 20% or more in a broad market index over at least a two-month period [? ?].

and new concepts coexist for some time, recurrent drift refers to settings in which previously observed concepts reappear after a period. Seasonal drift is a special case of recurrent drift where these recurrences follow a deterministic periodic pattern. For example, trading volume and volatility patterns may exhibit recurrent behavior tied to monthly options expiration cycles, quarterly earnings announcements, or annual tax-loss harvesting periods, with similar statistical properties recurring at regular intervals.

2.2. Statistical axis

The statistical axis specifies *which component* of the data-generating distribution changes over time. To understand, let $P_t(X, Y)$ denote the joint probability distribution governing the generation of input–output pairs at time t, with the factorization $P_t(X, Y) = P_t(Y \mid X)\, P_t(X)$. Here, $P_t(X)$ defines the probability distribution from which inputs are drawn, while $P_t(Y \mid X)$ defines the conditional probability governing the generation of outputs given the inputs.

In this taxonomy, categories are defined by the *dominant* changing term in this decomposition (or, equivalently, in $P_t(X \mid Y)\, P_t(Y)$), i.e., whether the drift primarily affects $P_t(X)$, $P_t(Y)$, $P_t(X \mid Y)$, or $P_t(Y \mid X)$. In practice, however, multiple components may evolve simultaneously. Accordingly, and consistent with the dominance-based definition above, the categories below should be interpreted as idealized descriptors that emphasize the primary source of change rather than as mutually exclusive cases.

- **Covariate / virtual drift:** change in $P_t(X)$ while the predictive mechanism $P_t(Y \mid X)$ is assumed *approximately invariant*. Here, "approximately" reflects an idealized assumption: empirically, one expects $P_t(Y \mid X) \approx P_{t'}(Y \mid X)$ *on the overlapping support of* X (i.e., regions where both periods assign non-negligible probability mass), up to estimation noise and minor residual effects. In practice, the drift is interpreted as being driven mainly by context/environment or sampling changes that move the distribution of inputs. For instance, a return forecasting model trained on a market index may face covariate drift when the composition of the index shifts toward technology companies and away from traditional manufacturing firms, even though the relationship between company-level features (valuation ratios, momentum, volatility) and expected returns remains stable.

- **Prior / label / target shift:** change in $P_t(Y)$ (class/event proportions, or the marginal distribution of the target). This captures situations where the frequency of outcomes changes over time, even if conditional relations are relatively stable. An example is a trading signal classifier where the proportion of buy/sell/hold signals changes over time due to changing market conditions, while the features that characterize each signal type remain similar.

- **Class-conditional shift:** change in $P_t(X \mid Y)$ with $P_t(Y)$ *approximately constant*. Again, approximately" indicates a working approximation: the marginal prevalence of classes/events is treated as stable (or controlled for) over the comparison window, while the feature distribution within each class moves (e.g., due to measurement, microstructure, or representation effects). For instance, in a model predicting order book price movements, the proportion of upward, downward, and stable price movements may remain constant, but the order flow patterns characterizing each movement class evolve as the composition of market participants shifts from predominantly retail to institutional traders, or as the prevalence of spoofing and fake orders changes over time. If $P_t(Y)$ also varies materially, the setting is better described as a mixture with prior/label shift.

- **Concept (strict sense) / real drift:** change in $P_t(Y \mid X)$, i.e., a change in the relationship between inputs and targets (often corresponding to "structural breaks" or "regime shifts" in econometrics/finance). For example, a price prediction model may observe that the same technical indicators (moving averages, volume patterns) that previously signaled upward price movements now predict downward movements, reflecting a fundamental change in market dynamics.

Many authors use the term *concept drift* in a broad sense, to denote any change in the data-generating process $\Delta P(X, Y)$, and not only real drift $\Delta P(Y \mid X)$ [**? ?**]. In this survey, we adopt the restricted convention: we reserve “concept drift” for $\Delta P(Y \mid X)$ (real drift) and use “distribution shift” or “non-stationarity” for the general case.

To avoid ambiguity, when cited authors use “concept drift” in the broad sense, we explicitly flag this and map it to our taxonomy by identifying which term in the standard factorizations of $P_t(X, Y)$ is drifting. Table 1 summarizes the main terminological equivalences across machine learning, econometrics, statistics, and quantitative finance for broad-sense "concept drift," mapping changes in $\Delta P_t(X)$, $\Delta P_t(Y)$, $\Delta P_t(Y \mid X)$, $\Delta P_t(X \mid Y)$, and $\Delta P_t(X, Y)$ to their commonly used names in each community.

In the dataset-shift literature, "dataset/distribution shift" typically refers to a mismatch between training and test distributions (i.e., the joint distribution differs across stages). In contrast, the concept-drift literature emphasizes online/streaming settings in which the distribution evolves over time, often affecting the input–target relationship.

Throughout this survey, we use "distribution/dataset shift" (or "non-stationarity") as an umbrella term that covers both offline (train–test) and online (time-varying) settings, while reserving "concept drift" for changes in $P_t(Y \mid X)$ under our restricted convention. [**? ?**]

But, in addition to these terminologies, other classic dimensions of non-stationarity can quantify how much and which statistical moments have changed[**? ? ? ? ?**]:

- **First order:** change in $\mathbb{E}[X_t]$ (mean, trend) or in univariate location statistics;
- **Second order:** change in $\mathrm{Var}(X_t)$ (volatility) or in $\mathrm{Cov}(X_t, X_{t-k})$ (autocorrelation)[2] [**? ?**]; and
- **Multivariate:** change in cross-dependence (e.g., covariance/correlation across variables or assets) and tail dependence [3] [**? ? ? ? ?**].

In multivariate settings, drifts may affect variances, covariance, correlation, and tail dependence across assets. Financial crises are typically marked by an abrupt increase in tail dependence, with losses converging across assets that were previously weakly correlated, leading to the collapse of diversification strategies (see Fig. 6). Conceptually, such changes can be described as breaks in covariance matrices, transitions in dependence graphs (financial networks), or changes in dynamic copulas that capture asymmetric dependence and heavy tails [**? ? ? ? ?**]. In these episodes, changes of different orders often coexist: mean shifts associated with flight-to-quality dynamics (Fig. 4), volatility explosions characterized by abrupt spikes in implied volatility (Fig. 5), and increases in correlations and tail dependence reflecting crisis contagion mechanisms (Fig. 6). These phenomena occur in parallel, combining first-order, second-order, and multivariate effects.

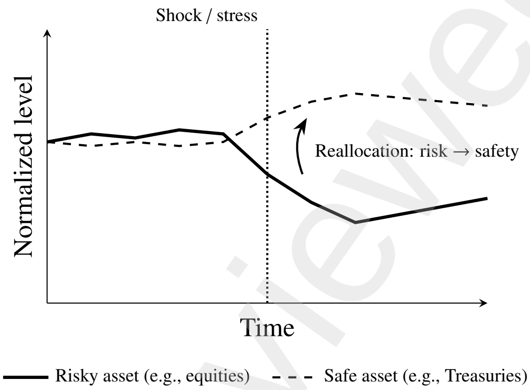

Figure 4: Schematic illustration of *flight-to-quality*: during stress, investors shift from risky assets to safer ones, depressing risky prices and supporting safe-asset prices.

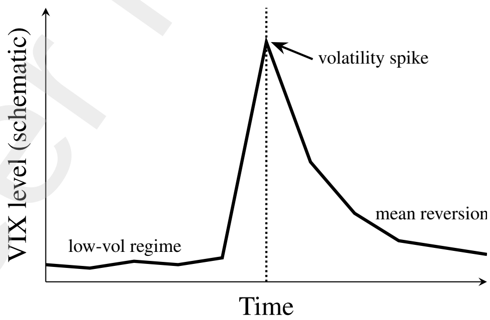

Figure 5: Schematic example of a *VIX spike*: an abrupt jump in implied volatility around a shock, followed by gradual normalization.

Figure 7 illustrates the typical inverse co-movement between equity prices and implied volatility: equity drawdowns often coincide with increases in VIX [**?**], which is computed from S&P 500 index option prices and is widely used as a market "fear gauge" [**?**]. As an example of a news-driven repricing episode, late-Jan. 2025 coverage reported a sharp tech-led selloff following DeepSeek-related developments, accompanied by a spike in volatility expectations [**? ? ?**].

In terms of the statistical axis, first/second-order tends to manifest as $\Delta P(X)$ changes in the features; $\Delta P(Y)$ affects marginal statistics of the target; and real drift $\Delta P(Y \mid X)$ manifests as instability of parame-

[2] For a univariate series $\{X_t\}$, the mean is $\mu_t = \mathbb{E}[X_t]$ and the autocovariance at lag h is $\gamma(h) = \mathrm{Cov}(X_{t+h}, X_t)$; the autocorrelation is the normalized quantity $\rho(h) = \gamma(h)/\gamma(0)$. In weak (second-order) stationarity, μ_t is constant and $\gamma(h)$ depends only on the lag h (not on t).

[3] Tail dependence captures extremal co-movement and is often quantified by coefficients defined as limits of conditional quantile exceedance probabilities.

[4] `https://github.com/davimcabral/NonStationarityInFinancialTS/blob/main/graphic_sp500_vix.ipynb`

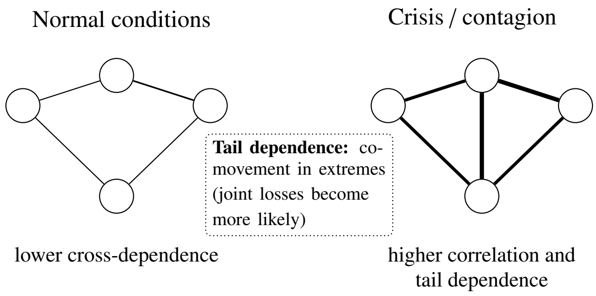

Figure 6: Schematic illustration of crisis contagion in a cross-asset network. Nodes represent assets (or variables) and edges denote statistical dependence, with thicker edges indicating stronger dependence. Under normal conditions (left), dependence is weaker and diversification is effective. During crises (right), dependence and tail dependence increase, making joint extreme losses more likely and reducing diversification benefits.

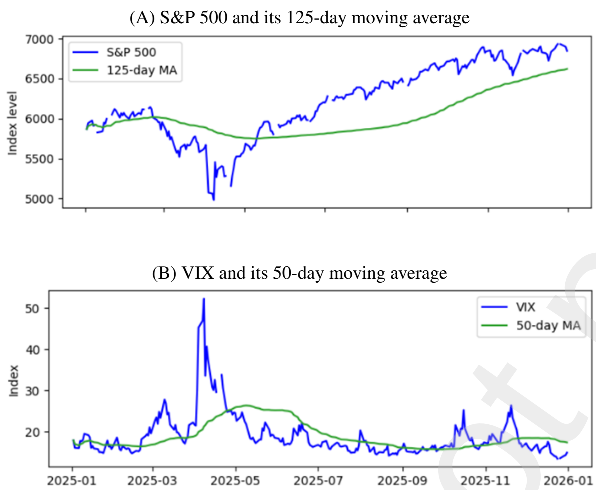

Figure 7: Co-movement between equity prices and implied volatility (daily data). *Authors' own plot.* The plotting notebook is available online.[4]. Data accessed via FRED (series SP500 and VIXCLS [**? ?**]; underlying index data are not redistributed).

ters/coefficients and of conditional dependencies, going beyond marginal moments [**? ? ? ? ? ?**].

2.3. Spatial axis

The spatial axis concerns the reach of a change within the feature space or across subpopulations, as illustrated in Fig. 8. It characterizes where in the input domain a change occurs and how broadly it spreads across variables, assets, or groups. From this perspective, spatial changes can be distinguished according to their extent, ranging from drifts that affect the entire domain to those confined to specific regions of the feature space, as formalized in [**? ? ?**], and described below:

- **Global drift**, when the change broadly affects the entire domain, impacting most regions of the feature space or the majority of subpopulations simultaneously. For instance, a central bank interest rate change typically affects pricing dynamics across all asset classes and market segments, inducing a system-wide shift in expected returns and risk premia.

- **Local drift**, when the change is confined to specific regions of the input space, such as a single economic sector, asset class, geographic market, or customer group. For example, regulatory changes affecting only the pharmaceutical sector may alter the predictive relationships for healthcare stocks while leaving technology or energy sectors unaffected. These changes are spatially heterogeneous, often subtle, and more difficult to detect, requiring region-aware or subdomain-sensitive monitoring methods.

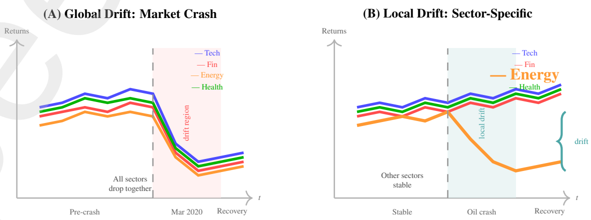

Figure 8: Spatial drift in financial time series. (A) **Global drift**: During the March 2020 COVID-19 market crash, all sectors experienced synchronized volatility increase and drawdown—drift affects the entire market. (B) **Local drift**: During the 2014 oil price collapse, only the energy sector (bold orange) experienced significant drift while other sectors remained stable. Local drift requires sector-specific detection and adaptation rather than market-wide retraining.

Beyond the distinction between global and local effects, spatial drifts may differ in their structural footprint across the feature space. Changes can propagate smoothly across neighboring regions, remain isolated within well-defined clusters, or emerge along specific dimensions while leaving others unaffected. Such patterns reflect the geometry of the input space and the interaction structure among variables, rather than temporal ordering, and are naturally described through partitions of the feature domain, network representations, or conditional submodels tailored to specific subpopulations.

Why do scales matter for the taxonomy? Drifts manifest at different frequencies, and the scale often determines which morphological pattern dominates. At macro scales (years/decades), drifts tend to be abrupt

Table 1: Terminological equivalences across traditions (broad-sense "concept drift").

	Phenomenon	Machine learning	Econometrics	Statistics	Quant finance
Concept Drift (broad sense)	$\Delta P(X)$	Covariateshift / covariate drift	Exogenous shift / exogenous shock	Covariate shift	Market dislocation
	$\Delta P(Y \mid X)$	Concept drift (strict sense)	Structural break	Conditional shift	Regime shift
	$\Delta P(Y)$	Label shift /prior-probability shift	Endogenous change	Target/ marginal shift	Behavioural shift
	$\Delta P(X, Y)$	Dataset shift / distribution shift	Non-stationarity	Joint shift	Market regime shift/ non-stationarity

(policy shocks, interest-rate regime changes; Fig. 3B) or incremental (secular trends linked to business and monetary-policy cycles;Fig. 3A) [**? ? ? ?**]. At daily/weekly scales, recurrent drifts prevail, associated with risk-on/risk-off switches, news cycles and repricing of risk around announcement windows [**? ? ? ?**]. At high frequency, intraday seasonality and microstructure effects (open/close patterns, auctions, lunch breaks) generate deterministic patterns that overlay structural drifts observed at longer horizons [**? ? ?**]. In crypto assets, boom–bust cycles tend to overlap with these scales, while deterministic events (such as halvings) act as natural triggers for regime changes [**?**]. Multiscale approaches [**? ? ?**] allow time series to be decomposed into different frequencies, reveal regimes that remain hidden at single resolutions, and thus classify drifts according to their dominant scale [**? ? ? ?**].

2.4. Ontological axis

Not all changes in data correspond to the same type of structural transformation. Some variations reflect transient fluctuations around a stable system configuration, while others signal a deeper change in how the data generation system operates. The ontological axis focuses on these qualitative differences between distinct system states.

A useful way to understand such structural changes is through the notion of a *regime*. Intuitively, a regime corresponds to a persistent mode of operation of the data generation system. While a regime is in place, data follow relatively stable patterns; when the regime changes, these patterns are altered in a systematic and lasting way. In financial markets, common examples include bull and bear phases, sustained transitions between low- and high-volatility states, and crisis periods characterized by persistently high cross-asset correlations.

In econometrics, from a modeling perspective, regimes are often treated as latent states that govern the data-generating process. This idea is formalized through piecewise models with approximate stationarity, where time series alternate between a finite number of discrete and persistent states separated by change-points (see Fig. 9). Transitions between regimes may be abrupt, as during financial crises, or gradual, reflecting slow structural adjustments in the economy.

This framework helps clarify the relationship between drift and regime change. Every regime transition necessarily involves a statistical change. But the converse does not hold, because many drifts arise from continuous or incremental adjustments that do not introduce a new regime. For instance, a brief spike in volatility represents a short-lived deviation, whereas a sustained period of elevated volatility over several months reflects a genuine regime change. The distinction between these cases is illustrated in Fig. 9.

In practice, a change is typically interpreted as a new regime when it satisfies three broad criteria [**? ?**]. First, the change must be persistent, lasting long enough to rule out transient noise. Second, it must be distinctive, meaning that its statistical properties differ meaningfully from those observed previously. Finally, regimes may be recurrent, in the sense that the same state can reappear over time, although recurrence is not a strict requirement.

Beyond regime changes, the ontological axis encompasses other forms of qualitative transformation in the data-generating process, capturing changes in what the system *is*, rather than only in how its statistical properties evolve. In financial applications, such transformations include shifts between persistent market states, the emergence of new market categories or instruments, and structural changes in how data are generated, as illustrated by examples summarized in Table 2.

These ontological changes intersect with the temporal, statistical, and spatial axes, but are distinguished by reflecting modifications in the underlying structure or semantics of the problem, rather than purely quantitative variation [**? ? ? ? ? ?**]. Common empirical drivers

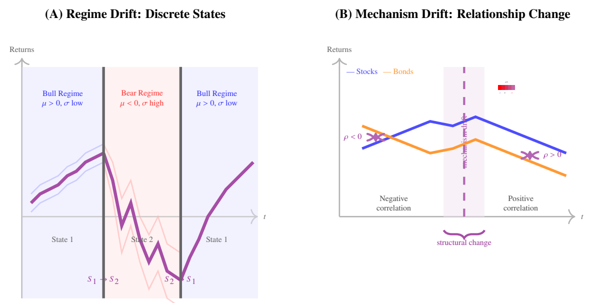

Figure 9: Ontological drift in financial time series. (A) **Regime drift**: Discrete state transitions between bull and bear markets. Each regime has distinct statistical properties (μ, σ), but the underlying data generation mechanism remains consistent within states. Modeled by Hidden Markov Models (HMMs), regime-switching models. (B) **Mechanism drift**: Fundamental relationship changes—stock-bond correlation shifts from negative to positive (as observed 2020-2022). The correlation structure $\rho(t)$ evolves continuously, representing a change in the causal/structural relationships rather than just state switching.

of these changes—such as policy interventions, technological innovations, or market disruptions—are further mapped to the corresponding axes in Table 3. *Within this framework, ontological changes can be organized into the following categories:*

- **Regime change** — transitions between persistent latent states with distinct statistical signatures, while preserving the same set of target classes (e.g., bull versus bear markets, low versus high volatility);

- **Concept evolution / new classes** — the emergence or disappearance of previously nonexistent target classes, effectively expanding or redefining the problem's ontology. For instance, the introduction of a new asset class (such as exchange-traded funds or cryptocurrency futures) creates novel prediction targets that did not exist in the training period, requiring models to recognize and adapt to entirely new categories;

- **Mechanism change** — modifications to the underlying generative process, such as the introduction of new variables, rules, or dependencies, even when observable classes remain unchanged. An example is the implementation of circuit breakers or new trading halts that alter market microstructure, changing how order flow translates into price movements without necessarily changing the classification targets (e.g., up/down/neutral price direction).

2.5. Causes of Non-Stationary

From a causal standpoint, drifts and regime changes can be triggered by exogenous, endogenous, or adversarial factors. Among exogenous drivers, monetary policy and interest-rate cycles, supply and demand shocks, financial crises, and geopolitical events stand out. These forces alter expectations, risk pricing, and risk premia across different time horizons [**? ? ? ?**]. Endogenous mechanisms arise from within the financial system itself. Leverage, liquidity constraints, margin calls, and feedback dynamics between investor strategies can amplify shocks and generate regime transitions even in the absence of new external events [**? ? ?**].

Table 3 summarizes how these causal drivers map onto the temporal, statistical, and spatial axes of the proposed taxonomy, illustrating how different sources of non-stationarity induce characteristic patterns of drift across dimensions. In addition, adversarial or manipulative behaviors—such as pump-and-dump schemes or spoofing practices in less liquid markets—may distort local signals. These actions often produce short-term drifts that do not correspond to long-term structural changes.

At high frequency, market microstructure effects play a central role. Latency, order flow, tick size, and market rules can create specific regimes, such as shallow or deep order books, narrow or wide spreads, and windows of elevated microstructural noise. These microstructure-driven regimes overlay macroeconomic regimes and contribute to the overall complexity of observed financial dynamics [**? ? ?**].

3. Representation and Context

The purpose of this section is to provide the foundations for addressing the following question: *how can financial time series be represented, and how can internal and external information be integrated, to support the detection and interpretation of regime changes?* For this, we adopt a layered approach to representation, illustrated in Fig 10.

At the first level, we consider internal signals derived from the time series itself, corresponding to endogenous information. At the second level, we incorporate exogenous context, representing external information that influences the observed dynamics. A third layer explicitly models the latent structure of underlying states or regimes as part of the representation, enabling a connection between observed data and unobserved market conditions. Fourth, we consider representations oriented

Table 2: Summary of drift taxonomic axes with financial examples

Temporal axis (how and when)	Statistical axis (what changes)	Spatial axis (where)	Ontological axis (structure/state)	Typical financial example
Abrupt	$\Delta P(Y \mid X)$	Global	Regime change (normal → crisis)	Market crash; jump in risk premium
Abrupt	$\Delta P(Y)$	Global	Same set of classes (no new ontology)	Sudden downgrade of sovereign or large-bank credit rating
Abrupt	$\Delta P(X)$	Local	Mechanism change (regulation / rules)	Regulatory shock in a specific sector
Gradual	$\Delta P(X)$	Local	Same set of classes (flows within the regime)	Slow sector rotation (out of one sector into another)
Incremental / continuous	1st/2nd order in $P(X)$	Global	Macro regime in slow drift	Secular downward trend in interest rates
Recurrent / seasonal	$P(X)$ / $P(Y)$ and 2nd order	Global or local	Recurrent seasonal regimes	January effect; intraday open/close patterns
Gradual	$\Delta P(Y \mid X)$	Local	Mechanism adjustment within the same regime	Market learning to price a new asset or technology
Abrupt	2nd order (dependence / tails)	Global	Contagion regime (normal → high dependence)	Correlation collapse in crises; loss of diversification
Abrupt	$\Delta P(X \mid Y)$	Local	Change in selection mechanism	New credit-scoring criterion changing the profile of approved clients

Table 3: Mapping drift causes to taxonomic axes

Cause	Temporal axis	Statistical axis	Spatial axis	Example
Monetary-policy shock (exogenous)	Abrupt	$\Delta P(Y \mid X) + \Delta P(X)$	Global	Surprise interest-rate hike
Liquidity contagion (endogenous)	Gradual	$\Delta P(X)$, 2nd order, multivariate	Global (systemic)	Cascading margin calls
Demographic trend (exogenous)	Incremental (long-run)	$\Delta P(Y)$	Global (country/region)	Population ageing
Pump-and-dump (adversarial)	Blip (transient)	$\Delta P(X)$, 1st order	Local	Manipulation in crypto markets
Bitcoin halving (exogenous)	Recurrent (deterministic)	$\Delta P(X) + \Delta P(Y \mid X)$	Global (crypto assets)	Supply-reduction events

toward robustness, emphasizing invariance and transferability across assets, time periods, and market environments.

The goal of these representations is to transform raw data into a feature space in which regime-relevant changes become explicit, whether as drifts in embeddings, reorganizations of network structures, or variations in latent indicators. Figure 10 provides a compact roadmap of how we organize drift-aware representations in this survey.

3.1. Internal Signals and Series Embeddings

Internal Signals. The most basic layer of representation consists of the internal signals contained in the time series itself — prices, returns, volumes, spreads, among others — analyzed through the lens of non-stationarity. Beyond classical handcrafted statistics, internal signals can also be summarized through learned representations. Fig. 11 illustrates this idea: each time window w_t is encoded into a latent vector z_t, so that drift can be monitored as a change in the trajectory $\{z_t\}$ in the embedding space.

Classical studies in time-series statistics and econometrics show that a large share of financial drifts manifests as changes in first- and second-order properties of the series (level, trend, volatility, autocorrelation), or in more complex forms of temporal dependence [**? ? ? ? ?**]. Thus, a natural starting point is to construct diagnostic features computed in moving time windows (or multiple time scales) that capture these statistics. Examples include:

- **Location and dispersion statistics:** moving means, medians, or quantiles; realized volatility in

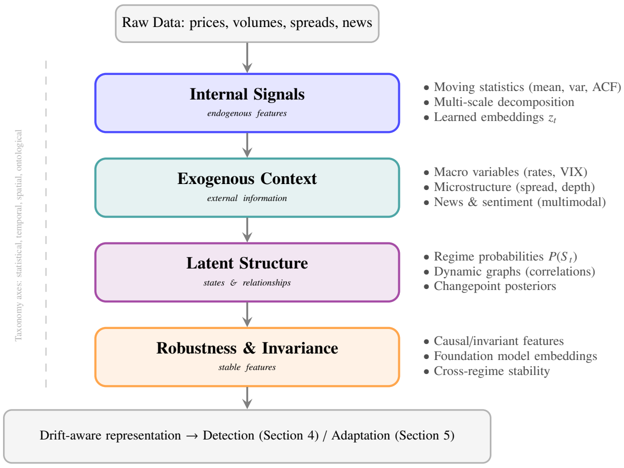

Figure 10: Representation layers for drift-aware financial modeling. Raw data is progressively transformed through internal signal extraction, exogenous context integration, latent structure modeling, and robustness-oriented features, producing representations that feed detection and adaptation systems.

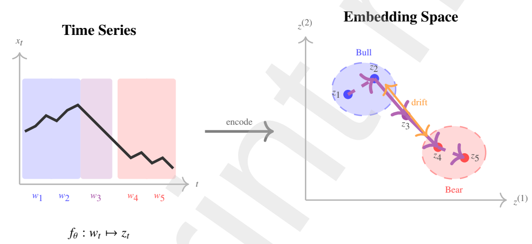

Figure 11: Learned embeddings for drift detection. Time windows w_t are encoded into latent vectors $z_t \in \mathbb{R}^d$. Similar market conditions cluster together (bull vs. bear regimes), while drift manifests as trajectory movement across the embedding space. Changepoint detection can be applied directly to the sequence $\{z_t\}$.

rolling windows; measures of skewness and kurtosis of returns.

- **Temporal dependence measures:** autocorrelation and partial autocorrelation coefficients (ACF/PACF [? ?]) at several lags; assessment of long memory via statistics such as long-run variance or unit-root tests (to identify changes between stationary and non-stationary trends).
- **Multi-scale indicators:** features obtained through wavelet decompositions or low-/high-pass filters at different horizons, to separate short-, medium-, and long-term movements [? ? ? ?]. For example, differences in means across scales (detection of trend changes) or energy metrics in specific frequencies (detection of emerging/vanishing cycles).

These features provide an aggregated view of how the statistical axis evolves when computed continuously in sliding windows. For example, one can monitor whether the mean or variance of returns exhibits a significant drift indicating a transition in the volatility regime [? ? ?]; track changes in the correlation between an asset and an important risk factor [? ? ?]; or detect instabilities in the estimated parameters of a local model (such as changes in the coefficients of a CAPM calibrated in rolling windows) [? ?]. Indeed, many change-detection approaches in financial time series rely on monitoring such statistics in real time. However, as the number of series and variables increases—such as when considering multiple assets simultaneously or multiple frequencies (e.g., daily and intraday)—relying solely on pre-defined, manually engineered features quickly becomes infeasible.

Series Embeddings. In this context, the concept-drift (broad sense) literature for data streams, together with advances in deep learning, suggests the need to *learn* representations automatically, rather than manually specifying all relevant statistics. The objective is to train a parametric function f_θ that maps sequences or time windows to dense vectors $z_t \in \mathbb{R}^d$ (i.e., temporal embeddings), such that periods exhibiting similar behavior are mapped to nearby regions in latent space [? ? ? ?]. This idea—learning embeddings that represent the latent state of the series—has become particularly prominent in non-stationary settings, as learned representations are often more effective at capturing complex drift patterns than hand-crafted indicators [? ?]. Table 4 summarizes the main drift-aware representation options discussed in this section, providing a structured overview before we examine specific approaches in more detail.

As an example, the TS2Vec approach proposed by Yue et al. [?] uses hierarchical contrastive learning to produce embeddings that are robust across multiple temporal resolutions. Essentially, the model is trained to generate consistent representations z_t for similar subsequences, while separating sequences with distinct patterns. Thus, locally similar windows remain close in latent space, whereas windows corresponding to different behavior patterns (e.g., bull vs. bear markets, low vs. high volatility periods) are mapped to distant regions of the embedding space. Studies report that such learned embeddings are useful not only for change detection but also for predictive tasks and anomaly detection in time series.

In the context of drift detection, operating in the latent space z_t has an advantage: classical changepoint methods can be applied directly to the embedding series[? ?]. Techniques such as CUSUM [? ?], PELT [?], or non-parametric (kernel) tests [? ? ?] can be run on sequences of vectors z_t instead of on the raw data. Since these embeddings are trained to condense high-dimensional relevant information (including nonlinear dynamics) into a few components, a significant change in series behavior tends to manifest as a detectable shift in the embeddings. In short, the learned representation "pre-processes" the data so that relevant changes are already highlighted, facilitating the detection stage.

A recent line of research goes further and explores training objectives oriented specifically toward anomalies and drifts. For example, contrastive learning methods that inject synthetic anomalies or perturbations during training, following a philosophy similar to exposing deep networks to outliers [?]. Here, embeddings are trained to maximize separation between "normal" and "altered" patterns. Such approaches include new-class detection methods in data streams [? ?] and self-supervised techniques like the CARLA model for time series, which uses contrastive objectives calibrated to highlight temporal anomalies [?]. The result is to bring the representation stage closer to the final detection task: one constructs a latent space designed to make breaks, novelty events, and out-of-distribution observations more evident [? ? ? ?]. In other words, instead of generic embeddings, one trains embeddings that are explicitly *sensitive to drift*.

Another important consideration is to embed the intrinsic multi-scale temporal structure of the data into the representation [? ?]. Different changes manifest at different time scales — for example, a secular trend versus intraday cycles — and choosing a single analysis scale *a priori* can lead to blind spots for certain types of change [? ?]. Models that integrate filters at multiple horizons, wavelet decompositions, or specialized layers for capturing distinct frequencies increase sensitivity to drifts at various scales [? ? ? ? ? ? ? ?]. Rather than deciding in advance on a single "correct" scale, these architectures internalize multiple frequencies in the representation. This connects directly to the temporal/morphological axis of the taxonomy (Section 2): by simultaneously considering long- and short-term components, we can detect both gradual long-term changes and seasonal or abrupt short-term shifts. Studies in high-frequency markets show that ignoring simultaneous scales — for example, analyzing only the macro trend and ignoring daily/intraday cycles — can lead to distorted assessments of volatility and regime identification [? ? ?]. Multi-scale embeddings therefore tend to provide robustness in the detection of complex changes.

3.2. Exogenous Context, Multimodality, and Market Structure

Macroeconomic and Exogenous Context. The second representation layer concerns the exogenous context that influences financial regimes. As discussed in Section 2.5, many drifts originate in macroeconomic shocks and cycles, monetary-policy decisions, liquidity crises, or geopolitical events [? ? ? ? ?]. Ignoring these contextual dimensions can lead to myopic representations that treat the series as an isolated system when, in reality, it responds to external factors.

A fundamental strategy is to incorporate macroeconomic and aggregate financial variables as covariates in the representation model. For example, one might include series such as short- and long-term interest rates, inflation indices, activity indicators, credit spreads, risk-aversion metrics (VIX, etc.), as well as established risk factors (value, momentum, size, etc.) [? ? ? ? ?]. These contextual variables can be integrated into the representation system in different ways:

1. **Direct feature fusion:** includes contextual variables as additional attributes concatenated to the internal features of the series in each time window. That is, the input to the representation model includes not only attributes derived from the target series, but also the corresponding values of macro indicators in that interval.
2. **State models with exogenous drivers:** in latent-regime models (such as HMMs[5] or MS-VAR [6]),

[5] A *Hidden Markov Model* (HMM) assumes a latent regime/state process S_t that evolves as a Markov chain, while observations Y_t are generated conditionally on the current state (via emission distributions). Inference typically relies on forward–backward recursions and EM/Baum–Welch-type estimation [? ? ?].

[6] A *Markov-Switching Vector Autoregression* (MS-VAR) is a VAR

Table 4: Comparison of embedding methods for drift-aware financial time-series representation.

Method	Learning Type	Real-time	Interpretability	Advantages	Limitations
Manual Statistical Features [? ? ? ? ?]	None (hand-crafted)	Yes	High (domain statistics)	Transparent; computationally light; domain knowledge integration	Requires feature engineering; limited to linear/simple patterns; doesn't scale to high dimensions
TS2Vec [?]	Self-supervised (contrastive)	Yes (after training)	Low (black-box)	Multi-resolution robust embeddings; no labels needed; captures complex patterns	Requires pre-training; limited interpretability; sensitive to augmentation design
Drift-Oriented Embeddings (CARLA) [?]	Self-supervised (anomaly contrastive)	Yes (after training)	Low–Medium	Explicitly sensitive to drifts; synthetic anomaly injection; tailored for detection	Requires careful anomaly design; may overfit to injected patterns; black-box
Regime-Switching (HMM/MS-VAR) [? ? ? ?]	Unsupervised (EM)	Yes (online filtering)	High (discrete regimes)	Probabilistic regime states; interpretable transitions; integrates macro drivers	Assumes discrete states; parametric assumptions; model misspecification risk
Bayesian Online Changepoint (BOCPD) [? ? ? ? ? ?]	Unsupervised (Bayesian)	Yes (online)	High (run-length posterior)	Real-time changepoint probabilities; uncertainty quantification; principled inference	Computational complexity; parametric likelihood assumptions; tuning priors
Multimodal (Series + Text) [? ? ?]	Supervised / Self-supervised	Yes (after training)	Medium (attention maps)	Early signals from news; rich context; captures narrative drivers	Alignment complexity; requires textual data; modality fusion design
Graph Neural Networks (GNN) [? ? ?]	Supervised / Self-supervised	Yes (after training)	Medium (graph structure)	Systemic view; captures cross-asset dependencies; network reorganization detection	Graph construction choice; scalability to large networks; dynamic edge computation
Foundation Models (Pre-trained) [? ?]	Self-supervised (transfer)	Yes (inference)	Low (black-box)	Generalization; data efficiency; leverages cross-domain patterns; minimal training	Domain gap; adaptation needed; update mechanisms; interpretability loss
Invariant/Causal Features [? ? ? ? ? ?]	Supervised (IRM)	Yes (after training)	High (causal structure)	Robustness to distribution shifts; focuses on stable relationships; reduces spurious correlations	Requires environment annotations; assumptions on causal graph; optimization complexity

macro variables can act as drivers that modulate transition dynamics between regimes. For example, interest-rate or inflation dynamics can be used as explanatory variables in the transition probabilities of a Markovian model [**? ? ? ?**]. This approach embeds context into the very state-change process, increasing interpretability (regimes can be associated with certain macro levels) and potentially improving the detection of transitions.

3. **Joint multimodal embeddings:** in deep architectures, one can train a network to learn joint representations that combine prices with macro indicators and other factors. In this case, rather than simply concatenating series, it is common to employ sub-networks or cross-attention mechanisms that integrate the modalities. The network then produces a unified embedding that simultaneously encodes the behavior of the financial series and its associated macroeconomic context [**? ?**].

Studies with regime-switching and MS-VAR models in finance suggest that using macro factors as part of the latent state improves regime separation and the economic interpretation of transitions. For example, a two-regime model can be much more interpretable if one regime is associated with "high inflation, rising rates" and the other with "low inflation, stable rates". Representing this context in the model state helps avoid spurious or practically meaningless regime detections.

Market Microstructure Context. The same principle applies at finer temporal resolutions, where the relevant context is no longer macroeconomic but structural. At high-frequency time scales, market dynamics are strongly shaped by market microstructure effects. Intraday behavior exhibits systematic patterns (opening, midday, closing, and auction periods) as well as distinct liquidity regimes throughout the trading day. As a result, representations designed for drift and regime detection at these horizons benefit from incorporating microstructure-related information, such as:

- **Order flow:** metrics of aggressiveness vs. passivity of orders; buy/sell imbalance in the order book;
- **Order-book indicators:** bid–ask spreads, depth (volume) available on each side, book changes;
- **Impact and resilience metrics:** how much the price moves after a large order (price impact) and how quickly the market recovers (liquidity resilience);

Together, these indicators make it possible to identify early signs of liquidity stress or structural changes in market functioning. Empirical studies show that such microstructure indicators can anticipate very short-term drifts related to phenomena such as flight-to-liquidity, liquidity crunches, or localized stress phases in specific markets [**? ? ?**]. That is, before a broader risk regime is established, we often observe deterioration in microstructure indicators (e.g., widening spreads, reduced depth) that signal a loss of liquidity. Incorporating these signals into the representation makes it reflect the current state of microstructure, which often precedes broader changes in price/risk regimes. In contrast, a representation based solely on aggregated prices may fail to capture these subtle structural changes.

Multimodal Context and Textual Information. In addition, an important aspect is the *multimodality* of information. Financial series rarely exist "alone": markets are constantly influenced and contextualized by textual information — newspaper articles, analyst reports, social-media posts, corporate and regulatory announcements, among others. Many disruptive events appear first as news or narratives before being fully reflected in prices. This motivates a class of multimodal representations that combine numerical time-series data with textual sources (and possibly other modalities, such as chart images, sentiment data, etc.).

Figure 12 provides an overview of the main multimodal fusion strategies, organized according to the stage at which information from different sources is integrated within the modeling pipeline. Depending on how modalities are combined, integration can occur at different processing stages:

- **Early fusion:** combine numerical and textual representations at the outset, for example, by concatenating series embeddings with news embeddings corresponding to the same time window. Each time window is then represented by a joint [$series + text$] vector.
- **Intermediate fusion (cross-attention):** use cross-attention mechanisms between time sequences and text sequences. For example, a transformer model in which relevant news influence — via attention — the learned representation of market data at that time. This allows the final series representation to

model whose parameters (e.g., intercept, autoregressive matrices, and often shock covariance) depend on a latent regime S_t governed by a Markov chain, allowing the multivariate dynamics to switch across regimes (e.g., low vs. high volatility) [**? ? ?**].

be modulated by textual content, amplifying or attenuating movements according to the presence of explanatory news.

- **Late fusion:** combine only at the end the outputs or alerts originating from detectors specialized in each modality. In this case, one could have a change detector operating on series and another analyzing news, and then merge detections (e.g., flag a change only if both agree, or use text as confirmation that a quantitative signal corresponds to a real event).

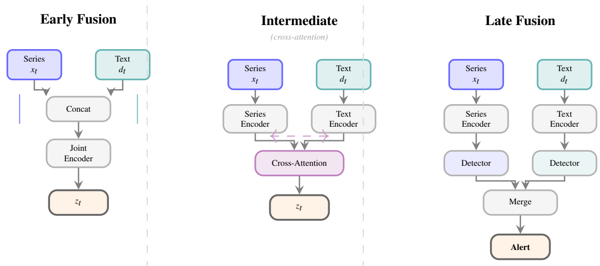

Figure 12: Multimodal fusion strategies for integrating time series and textual data. **Early fusion**: inputs concatenated before joint encoding. **Intermediate fusion**: separate encoders with cross-attention to exchange information. **Late fusion**: independent processing pipelines merged at the decision level.

These fusion strategies are not merely conceptual; they are increasingly adopted in recent research on multimodal time-series analysis and Large Language Models (LLMs). In particular, studies on applying LLMs to time series highlight architectures that perform temporal alignment and information fusion across numerical series, text, and other sources [**? ? ?**]. From the perspective of regime detection, the key advantage is that events described primarily in text (e.g., political or regulatory news that anticipates a market shift) can influence the joint representation before the corresponding movement is fully reflected in prices [**? ? ?**].

As a result, distances in multimodal space thus capture the combined effects of numerical and narrative signals: two periods will only be considered similar if both the quantitative patterns and the news context are similar. This enrichment can reduce false negatives (missing a change because the model did not "understand" that the news implied a new regime) and also false positives (distinguishing price drops caused by concrete events from those due to noise, via the presence or absence of textual explanations)[**? ? ?**].

Market Structure and Inter-Asset Dependencies. Beyond individual series and external signals, it is also important to account for the structure of interrelationships in the market as a dynamic graph. In multivariate settings, drifts rarely affect a single asset in isolation — typically, there is a reorganization of dependencies among assets. For example, in a given regime, economic sectors form clusters with high intra-cluster correlations and lower inter-cluster correlations; during crisis periods, correlations across most asset pairs tend to rise simultaneously and may approach unity, signaling a collapse of diversification and strong systemic contagion. Likewise, measures of tail dependence can change, revealing new channels of extreme co-movement [**? ? ? ? ?**]. Representing each series individually ignores this cross-information, whereas representing the system as an evolving graph allows us to capture co-movement regimes.

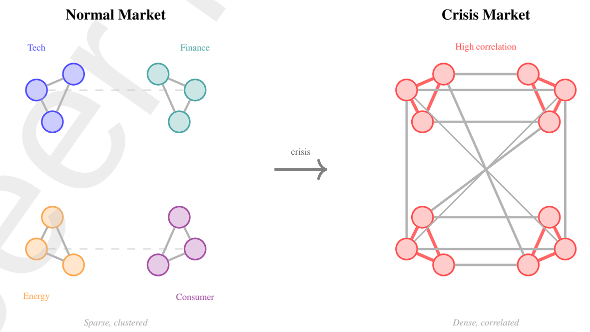

Figure 13: Dynamic correlation networks. **Normal market**: stocks cluster by sector with sparse inter-sector connections. **Crisis market**: correlations increase dramatically across all sectors (correlation collapse), with dense interconnection replacing the community structure. This regime change is captured by time-varying graph representations.

This intuition is illustrated in Figure 13, contrasting a sector-clustered correlation structure in normal periods with the dense, near-uniform dependence typically observed during crisis episodes. In such a graph, we typically define nodes representing financial entities (assets, indices, sectors, or countries) and weighted edges encoding some statistical or economic relationship among them. Edges may reflect, for example, estimated correlations or co-movements, measures of nonlinear dependence (tail copulas), exposure–credit relationships between institutions, or capital flows across markets. Edge weights vary over time, and regime changes appear as abrupt or gradual reconfigurations in network topology/weights. Examples include: during a financial-contagion event, sector clusters may dissolve and all nodes become highly interconnected; in subtler transitions, a new systemic hub may emerge (an asset or sector that starts to lead dynamics), or some links may weaken while others strengthen, reshaping the correlation network.

The recent literature on Graph Neural Networks (GNNs) for time series provides a framework for learning embeddings that encode such complex spatio-temporal dependencies [**? ?**]. Essentially, dynamic-graph models can learn both a representation for each node (asset) and a representation for the graph as a whole at each time interval. These network embeddings capture the current state of the financial structure — for example, they may encode that there are currently two weakly correlated clusters, or that a given asset is abnormally connected to others, indicating stress. Regime changes then manifest as changes in these embeddings: a shift indicating that the graph topology has changed significantly.

Recent financial models combine this graph-based approach with multimodality (prices + indicators + news), producing unified representations that account for relationships among assets together with economic context [**?**]. In practice, node and graph embeddings can serve directly as inputs to change detectors (Section 4) — for example, by monitoring the time series of the graph embedding to detect breaks in the correlation network — or as additional context for regime-based adaptation mechanisms (Section 5), guiding models to treat groups of affected assets jointly [**? ? ? ? ? ? ? ?**].

3.3. Representations Oriented Toward Robustness and Interpretation

This subsection introduces representation strategies oriented toward robustness and interpretability. The goal is to construct representations that remain stable under distribution shifts while providing economically meaningful signals of regime changes. A natural way to achieve this is to make regimes explicit in the representation itself, so that changes correspond to transitions between interpretable system states.

Latent-State and Regime-Based Representations.. One prominent class of approaches within this perspective relies on latent-state models with explicit regimes. Classical regime-switching models—such as HMMs, MS-VAR, or regime GARCH—represent the system as a finite set of discrete states, each associated with distinct statistical properties, and define transition probabilities among these states [**? ? ? ?**].

From a representation standpoint, fitting such a model amounts to mapping each time point to a vector of regime-membership probabilities, obtained either in real time (filtered) or a posteriori (smoothed). These vectors may be complemented by derived quantities, such as the predicted probability of remaining in the current regime or the expected regime duration.

In this sense, latent regime probabilities provide a compact and interpretable summary of the system state, integrating information from multiple variables into a small set of indicators. Figure 14 illustrates this idea: regime transitions appear as crossovers in the filtered probability paths, yielding drift-aware signals that are suitable for both detection and adaptation.

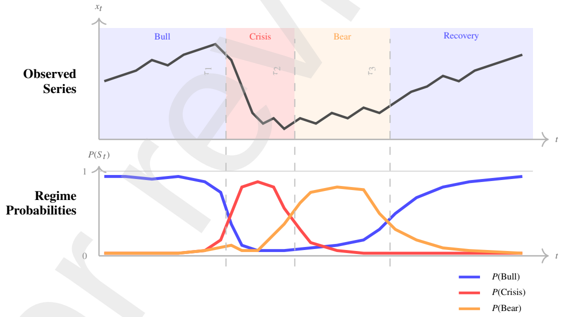

Figure 14: Regime probabilities from a latent-state model (e.g., HMM). Top: observed series with regime-colored background. Bottom: filtered probabilities $P(S_t \mid x_{1:t})$ for each regime. Transitions manifest as probability crossovers at changepoints τ_1, τ_2, τ_3. These probabilities serve as drift-aware features for detection and adaptation.

From a Bayesian standpoint, models such as Bayesian Online Changepoint Detection (BOCPD) and its non-parametric extensions provide an alternative yet related representation: they estimate, at each time t, the posterior distribution of the run-length (time elapsed since the last change) and of current model parameters [**? ? ? ? ? ?**]. In practice, BOCPD computes, at each new datum, a probability p_t(changepoint) indicating how likely it is that a change occurred at that point, and maintains parameter distributions under the no-change hypothesis.

The time series of these change probabilities and state statistics (for example, the posterior distribution of the currently active level or variance) become, themselves, rich representations of the degree of change perceived by the model. It is like a continuously updated "evidence panel": if p_t(changepoint) rises abruptly, we have a strong indication of drift; if uncertainty about parameters increases, this points to structural instability.

In summary, latent-state models (whether classical HMMs or online Bayesian methods) provide regime-oriented representations — mapping data into probabilities of scenarios — that both robustly capture relevant changes and enhance interpretability (since they make explicit "I am X% in regime A and Y% in regime B now").

Causal and Invariant Representations.. While this approach achieves robustness by explicitly modeling regime transitions, a complementary line of work seeks robustness at a deeper level: by identifying representations that remain valid across changing environments. This motivates a group of techniques focused on *causal and invariant* representations. The dataset-shift literature in classification emphasizes that many drifts arise from changes in confounders or in the surrounding environment—altering $P(X)$ or jointly $P(X, Y)$—while certain underlying causal relationships remain stable over time [**? ? ? ? ?**]. For example, an investment strategy may fail in a new regime because it relied on a spurious correlation that held historically but later vanished, even though the fundamental causal drivers of returns persisted (e.g., a macroeconomic factor whose effect was previously masked).

In response, a class of methods has emerged that aims to learn features invariant to environmental changes, drawing on ideas from Invariant Risk Minimization (IRM) [**?**] and related frameworks. Rather than optimizing only average predictive performance on historical data, these approaches impose constraints or regularizers that penalize excessive dependence on environment-specific components, encouraging the model to rely on more stable and transferable relationships [**? ? ? ?**].

In financial contexts, this amounts to seeking representations that privilege structural drivers (e.g., risk premia genuinely linked to economic fundamentals) and downweight correlations that, although effective in a given regime, are peculiar to that environment and do not persist outside it. As an example, imagine a representation of equities that emphasizes fundamental factors (valuation metrics, earnings growth, etc.) and is less sensitive to short-term technical factors whose sign may flip when the regime changes — such a representation tends to be more robust to market shifts, because fundamentals persist while transient technical patterns may disappear.

In practice, building invariant representations may involve: (i) selecting features whose relationship with returns remains stable across subperiods or regimes (identifying "resilient" variables whose estimated coefficients vary little between regimes); (ii) training embeddings with objectives that enforce predictive consistency across multiple environments (e.g., different market windows); (iii) integrating with latent-state models that make environments/regimes explicit and penalize excessive variation in causal effects across states. Recent studies show that causality-inspired models can yield more robust forecasts precisely in turbulent periods, when spurious relationships change and only structural links remain[**? ? ? ? ?**]. This suggests that invariant representations not only help detection (reducing noisy false alarms due to superficial variations) but also serve as a basis for adaptive models that maintain performance under drift.

Foundation Models and Universal Embeddings.. Along the same line of pursuing robust and transferable embeddings, recent work explores time-series *foundation models* as providers of universal embeddings. Inspired by the success of large pre-trained models in language and vision, researchers have examined whether large-scale time-series models, pre-trained on heterogeneous collections of series, can serve as backbones for diverse financial tasks [**? ?**]. The idea is that a neural model trained on a wide range of domains and frequencies learns embeddings of windows or entire series that carry temporal knowledge in a general way—capturing seasonal patterns, typical reactions to shocks, etc.—and that these embeddings can later be specialized to finance.

Instead of training a representation model from scratch on often limited market- or asset-specific data, one can adopt strategies such as: (1) using the pre-trained foundation model as a fixed feature extractor and applying change detectors to the embeddings it generates; or (2) performing light adaptation (fine-tuning), adjusting only the final layers to specialize the representation to the financial domain [**? ? ?**]. Both approaches aim to leverage the inductive bias contained in large-scale pretraining—for example, a foundation model may already "know" how to represent macro cycles or common shocks; even if the asset in question has never experienced a given regime in the available history, the model may recognize analogous patterns from other contexts and thus generalize better under moderate shifts.

From a pipeline perspective, embeddings provided by foundation models enter as enriched internal signals, representing the series through vectors of advanced features. This is a rapidly evolving area; although promising, it requires careful assessment of whether patterns learned from other domains apply to specific financial contexts, as well as mechanisms for updating the foundation model when the very "universe" of series evolves (for example, new data types or post-2020 dynamics). Well-designed representations not only increase sensitivity and timeliness in identifying changes, but also facilitate the economic interpretation of the resulting regimes—since they connect each drift alarm to understandable internal signals and potential structural mechanisms.

3.4. Design Guidelines for Drift-Aware Representations

Table 5 summarizes practical choices for representation and monitoring under different drift scenarios, linking observable drift patterns to concrete signal families and failure modes. In practice, representation design can be guided by a small number of recurring situations:

- *Abrupt shocks* (e.g., crashes, policy announcements): monitor short-window statistics (mean, variance) or regime probabilities from HMMs. These signals react quickly and are suitable for early warning, but may trigger false alarms under temporary noise.

- *Gradual or long-term changes* (e.g., persistent volatility increases): use longer rolling windows or multi-scale embeddings to track slow drifts, accepting later detection in exchange for stability.

- *Frequent regime switching*: adopt latent-state models (HMM, MS-VAR) and track regime probabilities or expected durations; performance depends on correctly specifying the number of regimes.

- *Asset- or sector-specific drift*: use dependency-based representations (graphs, tail dependence) to detect changes in correlations and contagion patterns, noting that estimates may become unstable in small samples.

- *Externally driven changes* (e.g., news, narratives): combine prices with textual data through multimodal embeddings to capture signals before they appear in returns, while accounting for alignment and scaling issues.

- *Need for robustness across market conditions*: favor invariant or causal representations to reduce reliance on regime-specific correlations, at the cost of discarding some short-term predictive signals.

Taken together, the checklist and table emphasize a simple principle: the representation should match the dominant form of drift one expects to face. Fast signals favor sensitivity, stable embeddings favor robustness, and interpretable regimes favor control and diagnosis.

4. Change Detection

This Section addresses the research question: *How can drifts in data be automatically detected over time?* To answer it, we organize the discussion around the main methodological paradigms used in practice. We first cover retrospective segmentation methods, which identify changes after observing a full data window (4.1). We then move to sequential (online) monitoring techniques, designed to trigger alarms as data arrive (4.2), followed by Bayesian approaches that model regime changes probabilistically (4.3). The section further examines detectors operating in learned representation spaces and out-of-distribution settings (4.4), as well as methods that target changes in multivariate dependence structures (4.5).

To synthesize these perspectives, 4.6 provides a compact overview that links the types of change defined by the taxonomic axes to suitable detection strategies. This overview serves as a practical guide for method selection, while also emphasizing how the choice of the observed signal and the representation space shapes detector behavior and interpretability [**? ? ?**]. Together, these elements connect methodological decisions to the effective detection of different forms of drift in real-world data.

4.1. Segmentation (retrospective optimization)

Segmentation methods seek to partition the historical time series $X_{1:T} = X_1, \dots, X_T$, where T denotes the total number of observations and X_t the observation at time t, into approximately stationary segments, retrospectively optimizing the changepoint locations that best explain the data. In this setting, it is assumed that the series is generated by a sequence of distinct regimes indexed by r, each associated with a probability distribution (or model) $P(X \mid \theta_r)$, where θ_r denotes the set of parameters characterizing regime r. Each regime is approximately stationary within its corresponding segment, and regime changes correspond to breaks in the underlying distribution or, equivalently, in the parameters θ_r.

In some cases, segmentation is applied not directly to X, but to representations $Z = f(X)$ or $Z = f(X, Y)$ (for example latent factors or features extracted from returns and covariates), so that one searches for breaks in $P(Z)$ over time. Segmentation algorithms then apply optimization routines (exhaustive or approximate) to find the optimal partition in terms of a global cost criterion [**? ? ?**].

A classical approach is the minimization of an additive cost criterion. One defines an intra-segment cost

Table 5: Design checklist linking drift taxonomy axes to representation choices.

Drift characteristic	**Signal / feature family**	**Early-warning indicator**	**Principal failure mode**
Abrupt drift	Short-window statistics, BOCPD, HMM probabilities	Sharp changes in mean, variance, or regime probability	False alarms due to transient noise
Gradual or secular drift	Multi-scale statistics, embeddings	Slow but persistent feature drift	Delayed detection
Regime switching	HMM / MS-VAR latent states	Probability crossovers, duration shifts	Regime misspecification
Localized or sectoral drift	Graph embeddings, tail dependence	Rewiring of dependency clusters	Estimation instability
Narrative-driven drift	Multimodal (price + text) embeddings	Text-led divergence before price moves	Alignment and modality noise
Spurious correlation drift	Invariant / causal representations	Stability across environments	Loss of short-term predictability

that usually coincides with a negative log-likelihood, for example

$$C(a, b) \approx -\log p(X_{a:b} \mid \hat{\theta}_{a:b}),$$

where $X_{a:b}$ is the data block in the segment and $\hat{\theta}_{a:b}$ are parameters estimated under the hypothesis that, in that interval, the distribution $P(X)$ (or $P(Z)$) is stationary. A cost is then added for each break introduced (a penalty as a function of the number of segments). Algorithms such as Bai & Perron [**?**] implement exhaustive search for multiple structural breaks by minimizing the total penalized cost.

Another more efficient method such as PELT (Pruned Exact Linear Time) [**?**], summarized in Algorithm 1, use a dynamic-programming recursion with pruning rules that exploit the additive structure of the cost function C to discard suboptimal candidates, achieving linear computational cost in many practical settings, with the penalty β controlling the number of detected changepoints.

Once the segmentation framework and optimization algorithms are defined, a key modeling choice concerns which aspect of the data-generating distribution is targeted by the cost function. Segmentation methods therefore require specifying whether changes are sought in the mean (location parameters), variance or covariance structure, or in the full distribution of $P(X)$ or $P(Z)$.

The choice of contrast statistic directly determines the type of changepoint that can be detected. Mean-based contrasts primarily identify level shifts, while variance-based contrasts are sensitive to changes in volatility. More general distributional contrasts, such as density-based measures, divergences, or energy

Algorithm 1: PELT: Pruned Exact Linear Time

Input: Time series $x_{1:n} = (x_1, \dots, x_n)$; cost function $C(\cdot)$; penalty β

Output: Set of changepoint locations $\mathcal{T} = \{\tau_1, \dots, \tau_K\}$

$F(0) \leftarrow -\beta$; `// Optimal cost up to time 0`
$\mathcal{R}_0 \leftarrow \{0\}$; `// Candidate set of last changepoints`
$\mathrm{cp}(0) \leftarrow \emptyset$; `// Changepoints up to time 0`
for $t = 1$ **to** n **do**
  `// Find optimal previous changepoint`
  $(F(t), \tau^*) \leftarrow \min_{\tau \in \mathcal{R}_{t-1}} [F(\tau) + C(x_{\tau+1:t}) + \beta]$
  $\mathrm{cp}(t) \leftarrow \mathrm{cp}(\tau^*) \cup \{\tau^*\}$; `// Store best configuration`
  `// Prune: remove candidates that cannot be optimal`
  $\mathcal{R}_t \leftarrow \{\tau \in \mathcal{R}_{t-1} \cup \{t\} \mid F(\tau) + C(x_{\tau+1:t}) \le F(t)\}$
end
$\mathcal{T} \leftarrow \mathrm{cp}(n) \setminus \{0\}$; `// Extract changepoints (exclude 0)`
return $\mathcal{T}$
Complexity: $O(n)$ average case with pruning, $O(n^2)$ worst case
Cost functions: C_{L2} (mean shift), C_{Gauss} (variance), C_{RBF} (nonparametric)
Penalty: $\beta = c \cdot \log n$ (BIC-type); higher $\beta \Rightarrow$ fewer changepoints
Typical use: Offline segmentation; multiple breaks; long series
References: [**? ?**]

distances, allow the detection of broader structural changes. In practice, segmentation often relies on simple within-segment models, including constant means, linear regressions, or AR/ARMA processes, with changepoints corresponding to shifts in the regime-specific parameters θ_r.

An important characteristic is that segmentation methods provide a set of estimated changepoints for the entire series (off-line mode), being suitable for exploratory analysis or historical validation of regimes. They do not operate in real time, but often offer high precision in *ex post* localization of significant changes in $P(X)$ or its parameters, especially when combined with robust stopping criteria (penalties) that avoid over-segmentation (inserting breaks where there is only noise) [**? ?**]. For example, techniques such as the Barry & Hartigan method [**?**] use partition models and priors for optimal segmentation with uncertainty, obtaining a posterior distribution over partitions of the series into approximately stationary blocks in terms of $P(X)$.

Beyond these canonical algorithms, there is a broad family of techniques that build upon segmentation as a basic block. Non-parametric extensions, such as E-Divisive, energy-distance segmentation, and kernel change point methods (kernel CPD), allow the detection of breaks in complex distributions $P(X)$ or $P(Z)$ without specifying an explicit parametric model, and are particularly useful for financial series with heavy tails and asymmetries.

Multi-scale methods based on random sub-intervals, such as Wild Binary Segmentation (WBS) and its extension WBS2, explore a large number of candidate segments to locate multiple changepoints, including scenarios with frequent breaks and very short spacing between drifts [**? ?**].

Narrowest-Over-Threshold (NOT), in turn, prioritizes the smallest interval whose contrast exceeds a threshold, producing well-localized estimates of features such as jumps in the mean or changes in slope and generalizing to different types of structural change [**?**].

In summary, segmentation methods frame changepoint detection as an offline optimization problem, in which a time series is retrospectively partitioned into approximately stationary regimes by minimizing a penalized global cost. Their effectiveness depends jointly on the choice of representation (X or Z), the intra-segment cost function, and the penalty controlling model complexity. As a result, segmentation provides a flexible and principled way to identify structural breaks and regime boundaries in historical data, serving as a foundational tool for exploratory analysis, regime characterization, and downstream modeling in non-stationary time series.

4.2. Sequential Methods (online monitoring)

Sequential methods perform detection in real time, examining the series continuously as new data arrive. In probabilistic terms, one typically assumes a "pre-change" distribution $P_0(X)$ and a "post-change" distribution $P_1(X)$ (or, more agnostically, one seeks to detect when $P_{\text{recent}}(X)$ starts to differ from $P_{\text{baseline}}(X)$). The goal is to trigger an alarm as quickly as possible after a change in $P(X)$ occurs, while controlling the false alarm rate. These methods typically maintain a window or adaptive model and apply recurrent statistical tests.

A classic example is the CUSUM (Cumulative Sum) test and its variants, detailed in the Algorithm 2, which monitor the accumulated deviation of a statistic (for example the mean of X_t or of a residual) relative to a reference value associated with $P_0(X)$, signaling a change when this deviation exceeds a threshold. The choice of the decision threshold and reference value (and the implied false-alarm/delay trade-off). In its likelihood-ratio formulation, CUSUM accumulates contrasts of the form $\log \frac{p_1(X_t)}{p_0(X_t)}$, approximating optimal detection between $P_0(X)$ and $P_1(X)$ under certain assumptions. Another example is the approach of Page–Hinkley, which states that derivatives follow a similar logic [**?**].

Another common formulation compares two windows: a recent sliding window, associated with an empirical distribution $P_{\text{recent}}(X)$, versus a past window or long-term estimate $P_{\text{baseline}}(X)$. Online two-sample test techniques, such as ADWIN and other drift detectors in data streams, continuously compute the statistical difference between these empirical distributions (e.g., in terms of mean, variance, or non-parametric measures of divergence) and perform sequential hypothesis tests, shrinking or expanding windows as needed to confirm a change in $P(X)$. Algorithms such as DDM, EDDM, etc., are used in the data-stream literature to monitor error metrics of a classifier over time – that is, an aggregated loss function $L_t = \ell(Y_t, \hat{Y}_t)$ when pairs (X_t, Y_t) are available – and trigger alarms when there is a significant increase in these metrics –indicating concept drift (strict sense) in $P(Y \mid X)$ [**? ? ?**].

A central challenge in these methods is controlling the trade-off between rapid detection and false alarms. Thresholds that are too sensitive lead to frequent alarms due to random fluctuations in $P(X)$; high thresholds are slow to react to real changes. For this reason, several techniques use results from sequential stopping theory: for example, defining thresholds that guarantee a certain level of ARL_0 (Average Run Length) for false detections – that is, on average, how long a change-free pe-

Algorithm 2: CUSUM for Mean Shift Detection

Input: Data stream $x_1, x_2, \ldots, x_n$; in-control mean μ_0; standard deviation σ; threshold h; slack parameter k (typically 0.5)
Output: Detection time τ or $\emptyset$ (no detection)

```
S⁺ ← 0 ;                // Cumulative sum for upward shifts
S⁻ ← 0 ;                // Cumulative sum for downward shifts
for t = 1 to n do
    z_t ← (x_t − μ_0)/σ ;          // Standardize observation
    S⁺ ← max(0, S⁺ + z_t − k) ;    // Update upward CUSUM
    S⁻ ← max(0, S⁻ − z_t − k) ;    // Update downward CUSUM
    if S⁺ > h or S⁻ > h then
        return τ = t ;             // Alarm: change detected
    end
end
return ∅ ;                         // No change detected
```

Complexity: $O(n)$ time, $O(1)$ space
Parameters: h controls ARL_0 (false alarm rate); k is the allowance for noise
Typical use: Abrupt shifts in mean; online monitoring; low latency
References: [**? ? ?**]

riod under $P_0(X)$ lasts until a false alarm occurs. Well-known sequential methods such as Shiryaev–Roberts or multivariate CUSUM calibrate thresholds via the desired ARL_0 [**? ?**].

Both approaches are necessary in financial applications, because online monitoring is used to detect structural changes in production: for example, warning that a risk model has started to fail because $P(X)$ or $P(X, Y)$ has changed. So that portfolio correlations, the distribution of P&L, or the rate of VaR violations are no longer compatible with the historical regime.

Some efficient implementations (for example, using low-complexity non-parametric tests) allow detectors to run in streaming on market data. It is worth noting that many sequential methods assume some knowledge of $P_0(X)$; in contrast, works on drift in data streams aim to be more agnostic, using sliding-window approaches and empirically comparing $P_{\text{recent}}(X)$ versus $P_{\text{baseline}}(X)$ or directly monitoring $P_{\text{recent}}(Y \mid X)$ through predictive performance.

In practice, sequential methods are the backbone of many operational tools in finance and machine learning for data streams. In risk monitoring, variants of CUSUM, EWMA, and GLR (generalized likelihood ratio tests) are applied to P&L series, counts of VaR violations, or volatility-model residuals to build early-warning dashboards that trigger limit reviews or stress tests whenever the statistic crosses pre-defined control bands, indicating that $P(X)$ or $P(Z = f(X))$ has changed.

In terms of algorithmic standpoint, sequential procedures based on GLR and SPRT, as well as control schemes on spread and liquidity indicators, act as "kill-switches" that halt strategies when recent behavior becomes incompatible with the historical regime. GLR compares online the likelihood of the data under no change against the best-fitting post-change alternative, while SPRT (Sequential Probability Ratio Test) accumulates evidence between two specified hypotheses until a decision threshold is reached. Control schemes based on spread and liquidity indicators follow the same logic.

In the supervised data-stream settings, libraries such as MOA, River, and Scikit-Multiflow provide DDM/EDDM/ECDD, ADWIN, KSWIN, and CUSUM variants as plug-and-play components within incremental classifiers, which in practice has consolidated these sequential detectors as de facto standards for monitoring covariate drift in $P(X)$, concept drift (strict sense) in $P(Y \mid X)$, and changes in time series in quasi-real time [**? ? ? ? ? ? ?**].

Finally, sequential detectors are often combined with segmentation methods: the online component provides fast, near-real-time alarms, while off-line segmentation refines the dating and statistical significance of structural breaks in subsequent analyses [? ?]. Together, these approaches balance responsiveness and statistical robustness, making sequential change detection a central building block in non-stationary time-series analysis.

4.3. Bayesian Methods

Bayesian methods model changepoint detection by specifying a probabilistic structure both for the occurrence of changes and for the data-generating process within each regime. Change times and regime states are treated as latent variables, jointly inferred with the model parameters.

It is assumed that the observations $X_{1:T}$—or pairs (X_t, Y_t) when responses are available—are generated from regime-dependent distributions $P(X \mid \theta_r)$ or $P(Y \mid X, \theta_r)$, where r indexes the latent regime and θ_r denotes its associated parameters. The temporal evolution of regimes is itself probabilistic, commonly modeled through hazard functions or Markovian dynamics.

Inference can be performed either sequentially (online) or retrospectively (off-line). As new observations arrive, or given the full data history, the model updates posterior distributions over the parameters θ_r and over latent quantities such as the run-length r_t (the time elapsed since the most recent change) and the probability of a changepoint at time t.

This framework naturally supports uncertainty quantification, allowing one to assess both whether a change occurred and when it occurred. It enables the principled incorporation of prior knowledge about regime persistence—such as preferences for rare changes or long-lasting regimes—through prior distributions on $P(\text{change at } t \mid r_t)$ and on the regime parameters $P(\theta_r)$.

Schematically, we can group these methods into three main subfamilies: (a) online detection via BOCPD and extensions; (b) retrospective Bayesian segmentation; and (c) regime-switching models (HMM/MS-VAR) with Bayesian filtering and smoothing.

BOCPD and online detection via run-length. In Bayesian Online Changepoint Detection (BOCPD) [?], the time series is modeled as a sequence of approximately stationary segments separated by latent changepoints, with the run-length r_t—the number of observations since the last changepoint—serving as the hidden state. At each new observation x_t, the algorithm recursively updates the posterior distribution

$$p(r_t \mid x_{1:t})$$

as summarized in Algorithm 3, via a message-passing scheme that combines: (i) the likelihood under the current regime, $p(x_t \mid \theta_{r_t})$ (or more generally $p(x_t \mid r_t, x_{t-r_t:t-1})$ for dependent models); and (ii) a prior on regime duration, encoded by a hazard function $h(r_t)$ specifying the probability of change as a function of r_t.

In terms of data distributions, each segment corresponds to a time interval in which $P(X)$ (or $P(Z = f(X))$) is assumed constant, and regime changes occur when the model deems it more likely that the recent data come from a new distribution.

At each step, the algorithm weighs two hypotheses: "continue in the current regime" (increasing r_t) versus "start a new regime now" (resetting $r_t = 0$). If the posterior probability of $r_t = 0$, $p(r_t = 0 \mid x_{1:t})$, increases sharply, this indicates a changepoint in near real time.

In finance, one often adjusts the hazard function to reflect beliefs about the frequency of regime breaks (for example, volatility shocks being rarer than smooth changes in the mean) and chooses likelihoods compatible with asymmetric or heteroscedastic returns, or even with distributions over representations $Z = f(X)$. Exact BOCPD has cost $O(T^2)$ in the length of the series, but run-length truncations or sliding windows allow $O(T)$ approximations.

Retrospective Bayesian segmentation (Barry–Hartigan, Fearnhead–Liu). A second line of approaches works in off-line mode, seeking the most probable segmentation (or samples from the segmentation distribution) given the entire history $X_{1:T}$ [? ?]. The change times $\tau_1, \ldots, \tau_K$ are explicitly modeled as latent variables, with prior distributions specifying both the number of segments K and the typical duration of regimes. In terms of data distribution, it is assumed that, within each segment k, an approximately stationary distribution generates the data, and that the task is to infer $P(\tau_{1:K}, K, \theta_{1:K} \mid X_{1:T})$.

Given this probabilistic formulation, inference can be performed either by maximum a posteriori (MAP) estimation, yielding a single "optimal" partition τ_k, or by MCMC-based sampling, which produces a distribution over possible segmentations and credibility intervals for each transition date. In both cases, the result is a Bayesian characterization of the regime structure of $P(X)$ over time, with uncertainty explicitly quantified.

This explicit treatment of uncertainty makes these models particularly appealing in financial applications, where the goal is often to reconstruct historical regimes ex post—such as bull and bear markets, high- and low-volatility periods, or episodes of policy intervention—while accounting for ambiguity in the precise

Algorithm 3: BOCPD: Bayesian Online Changepoint Detection

Input: Data stream $x_1, x_2, \ldots$; predictive likelihood $p(x_t \mid r_{t-1}, x_{1:t-1})$; hazard function $H(r)$; max run-length $R_{\max}$ (optional)
Output: Run-length posterior $p(r_t \mid x_{1:t})$ at each time t; changepoint probabilities

$p(r_0 = 0) \leftarrow 1$; `// Initialize: run-length is 0`
for $t = 1, 2, \ldots$ **do**
  `// Compute predictive probability for each run-length`
  for $r = 0$ **to** $\min(t-1, R_{\max})$ **do**
    $\pi_t^{(r)} \leftarrow p(x_t \mid r, x_{1:t-1})$; `// Likelihood under run` r
  end
  `// Growth probabilities: no changepoint at` t
  for $r = 1$ **to** $\min(t, R_{\max})$ **do**
    $p(r_t = r \mid x_{1:t}) \propto \pi_t^{(r-1)} \cdot p(r_{t-1} = r-1 \mid x_{1:t-1}) \cdot (1 - H(r-1))$
  end
  `// Changepoint probability: reset to` $r = 0$
  $p(r_t = 0 \mid x_{1:t}) \propto \sum_{r=0}^{\min(t-1,R_{\max})} \pi_t^{(r)} \cdot p(r_{t-1} = r \mid x_{1:t-1}) \cdot H(r)$
  `// Normalize posterior`
  $Z_t \leftarrow \sum_{r=0}^{\min(t,R_{\max})} p(r_t = r \mid x_{1:t})$
  $p(r_t \mid x_{1:t}) \leftarrow p(r_t \mid x_{1:t})/Z_t$
  `// Changepoint alarm (optional)`
  if $p(r_t = 0 \mid x_{1:t}) > \theta_{alarm}$ **then**
    Signal: Changepoint detected at time t
  end
  `// Update sufficient statistics for each run-length (model-dependent)`
  Update posterior parameters for $p(\theta_r \mid x_{1:t})$ *for each* r
end
Complexity: $O(n^2)$ without truncation; $O(nR_{\max})$ with truncation at $R_{\max}$
Hazard function: $H(r) = 1/\lambda$ (constant, geometric gaps); $H(r) = 1/(r+1)$ (increasing)
Likelihood: Conjugate pairs (Normal-Normal, Poisson-Gamma) enable closed-form updates
Typical use: Online detection with uncertainty quantification; gradual drifts; regime models
References: [? ? ?]

timing of regime boundaries for the relevant distribution $P(X)$ (e.g., returns, volatilities, or spreads).

Concrete instances of this class include the partition models of Barry–Hartigan [?], which place priors directly over partitions of the time series into approximately stationary blocks and derive the corresponding posterior distribution. Related algorithms developed by Fearnhead and collaborators [?] exploit dynamic programming or forward-type recursions to compute the joint distribution over the number and locations of changepoints, $P(\tau_{1:K} \mid X_{1:T})$. While exact inference typically incurs a computational cost of $O(T^2)$, resampling-based variants, such as particle filters, provide approximate solutions with near-$O(T)$ complexity.

Regime-switching models (HMM, MS-VAR). Finally, regime-switching models—most notably Hidden Markov Models (HMMs) and Markov-switching VARs (MS-VARs)—model regime changes through an unobserved discrete-state process $S_{t\,t=1}^{\;T}$. At each time t, the latent state $S_t \in 1, \ldots, R$ represents the active regime and evolves according to a first-order Markov chain, governing the model parameters in each period [? ?].

A classical example is Hamilton's model [?], which assumes that $X_t \mid S_t = r \sim P(X_t \mid \theta_r)$. In this setting, the series follows an autoregressive process whose coefficients and/or intercepts depend on the latent state, and each state r corresponds to a regime characterized by specific mean, volatility, and correlation patterns. In multivariate extensions (MS-VAR), $P(X_t \mid S_t = r)$ jointly captures regime-dependent autoregressive dynamics and covariance structures.

Inference in these models is typically Bayesian and combines filtering and smoothing procedures—such as the forward–backward algorithm—to estimate $p(S_t \mid x_{1:T})$ and $p(S_t, S_{t+1} \mid x_{1:T})$, together with parameter estimation via MCMC or variational methods. This yields posterior distributions for both the transition matrix $P(S_{t+1} \mid S_t)$ and the regime-specific parameters θ_r defining $P(X \mid S_t = r)$.

From a change detection perspective, regime-switching models can be employed online, by monitoring changes in state probabilities or transition likelihoods, and off-line, to reconstruct historical regimes and infer the most likely change dates. Their ability to jointly model multiple variables and complex dependence structures makes MS-VARs particularly well suited for multivariate financial series, where regime changes often involve simultaneous shifts in means, volatilities, and correlations—i.e., structural changes in $P(X)$.

In summary, Bayesian regime-switching methods provide a unified framework for regime detection and modeling, with explicit uncertainty quantification and the ability to incorporate prior information, such as assumptions about regime persistence or plausible parameter values. At the same time, their computational cost and sensitivity to modeling choices—such as the number of regimes, transition priors, and likelihood specification—require careful consideration, especially in high-dimensional financial settings.

4.4. Detection in Embeddings and OOD

This family of approaches detects changes by monitoring how data points evolve in a representation, or latent feature, space. Instead of operating directly on the raw observations, the idea is to first transform the series into a sequence of latent vectors and then track changes in their distribution over time. In practice, this is often framed as anomaly or out-of-distribution (OOD) detection in streaming settings.

Formally, a representation mapping $Z = f(X)$ or $Z = f(X, Y)$ is learned from the observed series X and, when available, from covariates Y. At each time t, a latent vector $z_t = f(x_{t-k:t})$ or $z_t = f(x_{t-k:t}, y_{t-k:t})$ summarizes recent observations through a sliding window. A regime change is then interpreted as a shift in the induced distribution $P(Z)$, which reflects an underlying change in $P(X)$ or $P(X, Y)$ [**? ? ? ? ?**].

A simple and intuitive example is provided by autoencoders or other unsupervised models trained on historical data. These models learn a notion of "normal" dynamics for the series. As new observations arrive, reconstruction errors or anomaly scores are computed. When these scores increase persistently, the current data no longer resemble the training distribution, indicating that the process generating X—and therefore Z—has changed.

Beyond reconstruction-based signals, changes can also be detected directly in the latent space. One approach is to define reference regions $P(Z)$ associated with known regimes, for instance, through contrastive learning or clustering. A regime transition is then detected when latent states drift away from these regions or move closer to others. Related methods compare distributions in representation space across time windows, using tools such as density-ratio or density-difference estimation, to explicitly test whether $P(Z)$ it has changed [**? ? ? ? ? ?**].

This general idea naturally extends to a wide range of deep OOD techniques. These include energy-based detectors, one-class and Deep-SVDD methods, generative models that estimate latent densities, and uncertainty monitoring in Bayesian or ensemble networks [**? ? ? ? ? ? ?**]. In multivariate financial applications, embeddings learned by transformers or graph neural networks are often combined with classical detectors—such as kNN, kernel density estimation, MMD tests, or standard CPD methods applied in latent space—to handle complex temporal and cross-sectional dependencies [**? ? ? ? ? ?**].

In finance, embedding- and OOD-based detectors are especially useful for identifying genuinely unprecedented situations, where patterns of co-movement, liquidity, or volatility deviate from all previously observed regimes. Their effectiveness, however, depends critically on the quality of the learned representation. When the embedding captures economically meaningful structure, changes in $P(Z)$ reliably signal changes in $P(X)$ or $P(X, Y)$. When it does not, these methods may confuse minor fluctuations with true novelty.

4.5. Multivariate Structural Dependence Methods

Finally, we highlight methods aimed at detecting regime changes that arise from shifts in the joint behavior of multiple variables. Instead of focusing on univariate marginals $P(X^{(i)})$ in isolation, these approaches operate directly on the multivariate distribution

$$P(X) = P(X^{(1)}, \dots, X^{(d)}),$$

Here $X_t = (X_t^{(1)}, \dots, X_t^{(d)})$ denotes the multivariate observation at time t, with each component $X_t^{(i)}$ representing a distinct variable of interest, such as the return of an asset, a risk factor, or a liquidity measure. Regime changes are then modeled as alterations in the dependence structure of the joint distribution, which may be reflected, for instance, in changes in the covariance matrix $\Sigma_t = \mathrm{Cov}(X_t)$, the correlation matrix R_t, a copula C_t linking the marginal distributions, or in the topology of a dependence graph (graphical model) associated with $P(X)$.

One example involves algorithms for detecting changes in the covariance matrix or in the graph of a graphical model. Typically, we compare P_{baseline} and $P_{\text{recent}}(X)$ via statistics that summarize their dependence: for example, tests for equality of covariance matrices Σ_{baseline} vs. Σ_{recent}, or statistics of maximal difference in correlation coefficients ρ_{ij} over time. In this direction, methods such as ICSS [**?**], MOSUM [**?**], and PELT extensions for the covariance matrix (PELT–Σ) [**?**] monitor breaks in univariate variance and, in multivariate versions, in joint variance/covariance, being useful for identifying volatility and co-movement regime changes [**? ?**].

Building on this idea, several methodological lines have been developed. On the sequential side, multivariate versions of the CUSUM of squares [? ?] monitor sums of squares (or aggregated portfolio statistics) over time to detect changes in joint volatility [? ?]. Another line focuses on techniques for detecting changes in copulas C_t, which describe exclusively the dependence structure between variables, decoupled from marginals F_i. In this case, the idea is to compare an old" copula C_{baseline} with a new" copula C_{recent} and test whether there has been a change, including in tail regions [? ? ? ? ?].

Some parametric methods assume Gaussian graphical models or related structures. An example is the use of (possibly adaptive) Graphical Lasso to estimate, in each window, a precision matrix $\Theta_t = \Sigma_t^{-1}$ associated with a conditional independence graph G_t [?]. In this context, monitoring changes in dependence amounts to monitoring changes in G_t or Θ_t.

Other methods avoid strong parametric assumptions and compare multivariate dependence measures constructed directly from the data, such as distance matrices $D(X)$ or kernel matrices $K(X)$ [?], using multivariate Friedman–Rafsky-type tests [?] or graph-based variants (MST, k-NN graph) [? ?] to assess whether two samples originate from the same joint distribution $P(X)$.

On the Bayesian side, BOCPD extensions tailored to detecting changes in structural dependence have also been proposed [? ? ? ? ?]. In these models, the dependence structure (for example, a graph G or a parameter set Θ encoding the edges) is treated as latent and evolving by regimes, and inference focuses on detecting abrupt switches in this structure over time.

From a structural standpoint, methods based on correlation matrices, copulas, or graphical models interpret these events as rearrangements in the dependence network [? ? ? ?]. This perspective is particularly relevant in financial applications, where structural changes tend to manifest more strongly in the relationships between variables than in each series taken individually. In distributional terms, this means that changes in $P(X)$ are often driven primarily by shifts in its dependence component, while the marginals $P(X^{(i)})$ may remain relatively stable.

Importantly, such changes can often be detected before any univariate model signals instability, since individual series may remain within typical variation ranges while the joint dependence structure becomes unprecedented [? ? ? ? ?]. For instance, a contagion regime may not be evident when inspecting individual asset returns, but it becomes clear when correlations rise collectively or when the tail copula concentrates more mass on joint extreme events. Similarly, historically stable hedge relationships may deteriorate or invert sign, translating into localized changes in Σ_t and R_t and into increased residual volatility of long–short strategies.

4.6. Change Detection Overview

This overview is anchored by three complementary summaries. Together, Table 6, Figure 15, and Table 7 map the type of change of interest to suitable detection methods, practical selection criteria, and their computational feasibility.

Table 6 serves as the primary entry point. It maps different change axes—temporal, statistical, structural, and ontological—to indicative method families and concrete financial examples. For instance, abrupt temporal changes suggest segmentation or CUSUM-type methods, while shifts in dependence structures point toward covariance-, copula-, or graph-based detectors. This compact view allows the practitioner to start from the phenomenon of interest (e.g., contagion, regime changes, beta instability) and narrow the methodological space accordingly.

Figure 15 operationalizes this mapping as a decision tree. Starting from label availability, it guides method selection through a sequence of practical questions about dimensionality, operational objective, usage context (online vs. offline), and scale of concern. In this way, the figure connects conceptual choices to implementable detector families, highlighting trade-offs between supervision, sensitivity, and interpretability.

A complementary decision concerns the *representation space* in which change is monitored. As summarized in Table 6 and operationalized in Figure 15, detectors may act on raw series, on hand-crafted financial features, or on learned embeddings, depending on dimensionality and interpretability requirements. This choice determines whether changes are detected directly in observable quantities (e.g., returns or correlations) or indirectly through shifts in latent multivariate patterns.

Finally, Table 7 constrains these choices from a computational perspective. It highlights how method families differ in time and memory complexity as a function of series length, dimensionality, and model structure. This comparison is essential in high-frequency or high-dimensional settings, where theoretically appealing methods may be impractical without dimensionality reduction or truncation strategies.

Taken together, these complements transform the broad taxonomy of change detection into a practical selection framework: from identifying the relevant change axis, to choosing an appropriate method family, representation space, and control mechanism, all while respecting computational constraints.

5. Adaptation and Continual Learning

This section addresses the research question: *How can we adapt model learning to data distribution shifts continuously and effectively?* The focus is on strategies that enable learning systems to react to evolving data distributions in order to sustain or improve performance in non-stationary environments, with particular relevance to financial applications [**? ?**].

The discussion is organized around the main paradigms for continuous adaptation. We begin with parametric adaptation methods that incorporate change mechanisms directly into statistical models (5.1). We then examine dynamic ensembles and regime-specialized models that adaptively activate or reweight learners across regimes (5.2). Next, we cover hybrid adaptation flows that combine explicit change handling with model updating (5.3). Finally, we discuss recent approaches based on continual learning, test-time adaptation, and meta-learning, which aim to enable rapid and data-efficient adaptation (5.4) [**? ?**].

To synthesize these perspectives, Table 8 provides a structured comparison of adaptation methods across key operational dimensions, serving as a practical guide for method selection under different application constraints. This synthesis highlights the central trade-off between stability and plasticity that characterizes adaptive learning in finance: overly aggressive adaptation may amplify noise and transaction costs, whereas conservative updates risk sustained performance degradation after regime changes [**? ?**]. Together, these elements connect adaptation strategies to their practical implications in real-world non-stationary systems.

5.1. Parametric adaptation approaches

Within this family, we distinguish three main parametric strategies according to how change is represented in the model. The first relies on continuously evolving parameters, typically formulated in state-space form (5.1.1). The second introduces regime dependence through observable drivers via threshold or smooth-transition mechanisms (5.1.2). The third assumes a finite set of discrete latent regimes, with probabilistic switching dynamics captured by regime-switching and Hidden Markov models (5.1.3).

5.1.1. Time-varying parameters (TVP / state-space)

Time-varying-parameter (TVP) models address non-stationarity by allowing model coefficients themselves to change over time. Instead of estimating a single fixed parameter vector, the model treats coefficients as evolving quantities that are updated as new data arrive, typically through a state-space formulation [**? ? ?**]. This perspective naturally leads to regression models in which parameters follow an explicit evolution equation.

$$y_t = \beta_t^\top x_t + \varepsilon_t,$$

in which β_t follows an evolution equation $\beta_t = \beta_{t-1} + u_t$ (a random walk, possibly with some structure). In finance, this is applied to dynamic market betas (betas that change over time) and other adaptive regression coefficients.

Estimation is typically carried out using the Kalman filter or its extensions (for linear-Gaussian cases) or through online optimization methods for more general cases [**? ? ?**]. The Kalman filter directly provides recursive coefficient updates as new data arrive, using a transition model that penalizes overly abrupt changes (through the variance matrix Q of the term u_t). This yields an optimized adaptive forgetting: if the data indicate that β is changing, the filter adjusts; if not, it keeps it stable. In many cases, this "smooth" layer of adaptability responds adequately to incremental or gradual drifts, reserving more radical interventions (such as resets or model switches) for moments of clear structural break.

For example, suppose a stock's sensitivity to a market factor slowly increases over several months — a TVP model with Kalman filtering will gradually raise the estimated beta, tracking the change without ever having to explicitly declare a one-shot "regime change". This avoids losing accumulated information and ensures smoothness. However, if an abrupt shock drastically changes β, the TVP model will take a few steps to fully adjust (unless one temporarily increases the transition variance Q at that instant, which is equivalent to detecting and nearly resetting — hence the interaction with detectors).

In summary, TVP and state-space models provide a form of "built-in continuous adaptation," especially useful when we expect parameters to move slowly. In financial time series, there is broad application: stochastic-volatility models with time-varying parameters, macro VAR models with varying coefficients, dynamic CAPM models, and so on [**? ?**]. They tend to preserve economic interpretability (e.g., one can track how a given coefficient evolves and relate it to market conditions).

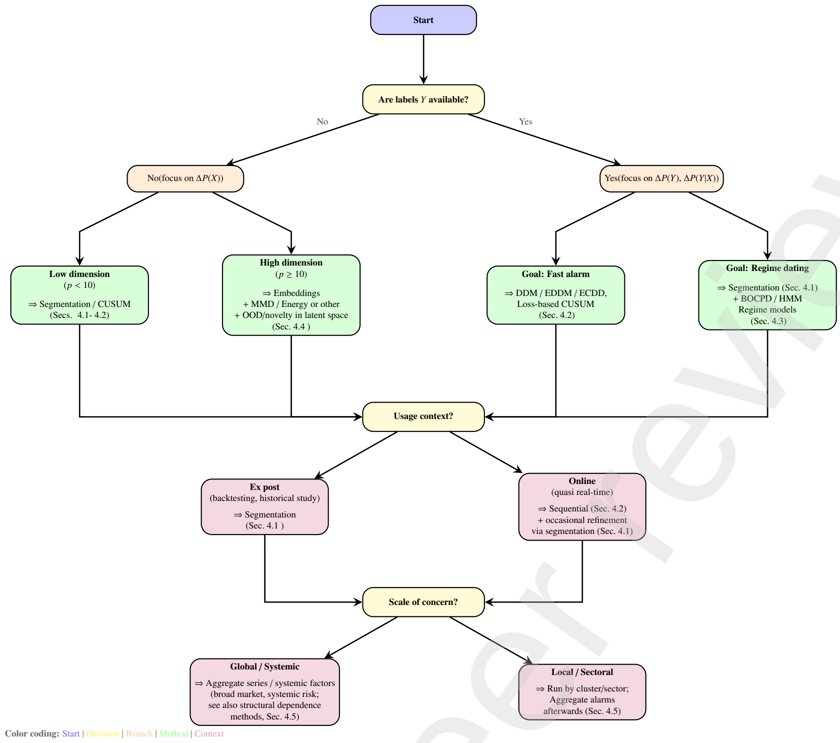

Figure 15: **Decision tree for choosing change-detection method families.** The flowchart guides method selection based on label availability, dimensionality, operational goal, usage context, and scale of concern. Colors distinguish decision nodes (yellow), branching points (orange), recommended methods (green), and contextual considerations (purple).

5.1.2. *Smooth-transition and threshold models (TAR/STAR)*

In some applications, changes in behavior are associated with observable variables that indicate when the system is operating under different conditions. Threshold and smooth-transition models build on this idea by allowing model parameters to change as a function of such variables, rather than evolving autonomously over time. This results in adaptation mechanisms in which regime changes are triggered by explicit drivers.

In TAR models, one defines one or more thresholds on a state variable (which may be the series itself at a lag, or another variable) that determine discrete changes in the parameters. For example, a bilinear TAR for returns: if the variable z_{t-d} (which may be a lagged return or a macro indicator) is below a threshold γ, we use one set of parameters (μ_1, ϕ_1, etc.); if it is above, we use another set (μ_2, ϕ_2).

This results in deterministic regime switching: not stochastic as in an HMM, but governed by the driver z. A financial example is a price-momentum model in which, if a short-term return exceeds a given threshold, the system enters a high-volatility regime or a regime with different mean-reversion dynamics.

STAR (Smooth Transition Autoregressive) models, in turn, implement the transition in a continuous way: parameters are weighted combinations of two (or more) base regimes via a smooth function (typically logistic) of the driver z_t [**? ? ? ?**]. Thus, if z_t is in extreme values, the model approximates a pure regime; in mid-range values, it is a mixture of the two. This is useful for capturing gradual changes or situations in which the regime does not switch abruptly but as some indicator deteriorates or improves. For example, a STAR model for inflation: as expected inflation gradually moves from X% to Y%, the monetary-policy regime (central bank reaction parameters) transitions smoothly.

These models allow us to incorporate, ex ante, knowledge about which variable signals a behavioral change. In finance, there are clear cases: for instance, a volatility model whose regime depends on an implied-volatility indicator (VIX [**?**]) — when VIX is high, different parameters apply. Or a consumer-credit model that changes when the unemployment rate exceeds a given threshold. By specifying this, adaptation becomes automated: the model instantly adjusts its parameters when

Table 6: Change axes and indicative methods (compact view).

Axis / feature	**Typical methods**	**Illustrative financial example**
Temporal – abrupt	Segmentation (PELT, Bai–Perron); CUSUM	Sudden crash in a market index
Temporal – gradual	EWMA/GLR; BOCPD with smooth hazard	Slow transition from bull to bear regime
Statistical – $\Delta P(X)$	E-Divisive, kernel CPD; ADWIN/KSWIN	Sector rotation in stock returns
Statistical – $\Delta P(Y \mid X)$	Regression segmentation; supervised CUSUM	Instability of betas in a factor model
Structure – Σ / dependence	PELT–Σ; CUSUM of squares; copula tests	Increase in correlations during crises (contagion)
Ontological – new classes/regimes	BOCPD/HMM (recurrent regimes); OOD in embeddings	New market regimes; unprecedented events

the driver crosses the threshold (TAR), or adjusts gradually (STAR), without needing to be re-estimated from scratch.

The limitation is obvious: one must choose the driver and calibrate the threshold/transition function. If the cause of regime change is not clear, these models may not be applicable. However, when the driver is well chosen, they offer rapid adaptation (virtually without delay if the driver is observed in real time) and usually stability within each regime, since each submodel can be calibrated for that context.

5.1.3. *Discrete latent states (HMM / regime-switching)*

Another classical way of modeling structural change is to assume that the data-generating process switches between a small number of distinct regimes over time. In this approach, regime membership is not directly observed but inferred from the data, and each regime is associated with its own set of parameters. This assumption underlies regime-switching and Hidden Markov models [**? ? ? ?**]. Regime-switching models with discrete latent states are arguably the classical parametric approach to structural breaks.

In this framework, a latent discrete-valued process $S_t \in \{1, \dots, K\}$ indicates the regime active at time t. Conditional on S_t, the observed variable is generated by a regime-specific distribution, with each regime characterized by its own parameters (e.g., means, variances, or coefficients). This allows the data-generating process to switch abruptly between distinct parameterizations.

A Hidden Markov Model (HMM) specifies the regime dynamics by assuming that $\{S_t\}$ follows a first-order Markov chain with transition probabilities

$$P(S_t = j \mid S_{t-1} = i) = \Pi_{ij},$$

where Π encodes regime persistence and switching behavior [**? ?**]. Together with the regime-conditional observation models, this transition structure defines the joint evolution of latent states and observations.

Model parameters and latent-state probabilities are typically inferred using the Expectation–Maximization (EM) algorithm or Bayesian methods such as MCMC [**? ? ?**]. In finance, these models are widely applied, for example in Markov-switching GARCH and VAR formulations [**? ? ? ?**], where volatility or mean levels can shift abruptly across regimes.

In the adaptation context, latent-regime models act proactively: they incorporate regime change in the model structure itself. When a new regime occurs, the model recognizes it (via filtered probabilities) and switches to the corresponding parameter set, which may differ radically from the previous one. Thus, we avoid "forcing" a single parameter set across the entire history.

However, a classical HMM with a fixed number of regimes K has adaptive limitations: it can only alternate among those K pre-estimated patterns. If a qualitatively new regime emerges, the model does not explicitly represent it (unless K was chosen larger and that regime occupies one of the slots). In other words, it handles recurrences of known regimes well but not truly novel regimes.

Even so, within the set of modeled regimes, HMM adaptation is fast — as soon as the filtered probability of a new state exceeds, say, 0.5, the model essentially uses that state's parameters, which is much faster than recalibrating a model from scratch. Moreover, the Markov structure imposes some inertia (e.g., if Π_{ii} is high, states are persistent, and the model does not immediately switch back), which prevents reacting to each fluctuation.

5.2. *Dynamic ensembles and regime specialization*

No single model performs well across all regimes, particularly in non-stationary environments where

Table 7: Computational complexity of change-detection methods. n = series length, p = dimension, S = number of states (HMM), $R_{\max}$ = run-length truncation (BOCPD), K = number of changepoints.

Method	Time	Space	Real-time?	Scalability	Notes
Sequential Methods (Section 4.2)					
CUSUM	$O(n)$	$O(1)$	Yes	High	Single pass, constant memory
Page-Hinkley	$O(n)$	$O(1)$	Yes	High	Variant of CUSUM
EWMA	$O(n)$	$O(1)$	Yes	High	Exponential smoothing
DDM/EDDM	$O(n)$	$O(w)$	Yes	High	Window size w
ADWIN	$O(n \log n)$	$O(\log n)$	Yes	Medium	Dynamic window
Segmentation Methods (Section 4.1)					
Binary Seg.	$O(n \log n)$	$O(K)$	No	High	Greedy approximation
PELT	$O(n)$ avg $O(n^2)$ worst	$O(n)$	No*	Medium	$O(n^2)$ worst; pruning critical
WBS	$O(n \log^2 n)$	$O(n)$	No	Medium	Wild binary segmentation
Bai-Perron	$O(Kn^2)$	$O(n)$	No	Low	Regression breaks
Bayesian Methods (Section 4.3)					
BOCPD	$O(n^2)$ $O(nR_{\max})$ trunc	$O(n)$ $O(R_{\max})$	Yes*	Low	$O(nR_{\max})$ with truncation
HMM (Forward)	$O(nS^2)$	$O(S)$	Yes	Medium	Online filtering
HMM (Viterbi)	$O(nS^2)$	$O(nS)$	No	Medium	MAP sequence
HMM (Baum-Welch)	$O(TnS^2)$	$O(nS)$	No	Low	T EM iterations
Embedding Methods (Section 4.4)					
Embed + CUSUM	$O(np + p^3)$	$O(np)$	Yes	Medium	p^3 from embedding training
MMD (kernel)	$O(n^2 m^2)$	$O(nm)$	No	Low	Windows n, m; kernel matrix
Energy Distance	$O(nm \log nm)$	$O(nm)$	No	Medium	Sorting-based
OOD (density)	$O(np + p^2)$	$O(p^2)$	Yes	Medium	After embedding trained
Structural Dependence (Section 4.5)					
ICSS (Σ)	$O(np^2)$	$O(p^2)$	Quasi	Medium	Covariance breaks
CUSUM of squares	$O(np^2)$	$O(p^2)$	Yes	Medium	Multivariate variance
Copula tests	$O(n^2 p^2)$	$O(np)$	No	Low	Tail dependence

Scalability (rule-of-thumb): ratings indicate the typical maximum series length n that can be handled on a standard workstation with an efficient implementation, *assuming small-to-moderate dimension* ($p \lesssim 20$). High: $n \gtrsim 10^6$; Medium: $10^5 \lesssim n \lesssim 10^6$; Low: $n \lesssim 10^5$. In practice, large p and expensive operations (e.g., kernel matrices) reduce these limits substantially.
*** Real-time note:** PELT is offline, but can be used in a quasi-online way via sliding/rolling windows. BOCPD is online, but the naive implementation is $O(n^2)$; real-time operation typically uses run-length truncation to a maximum $R_{\max}$ (e.g., 100–500), yielding $O(nR_{\max})$ time and $O(R_{\max})$ memory.
Practical considerations: for high-dimensional streams ($p > 100$), dimensionality reduction (e.g., PCA or learned embeddings) is usually required before applying most detectors. Sequential methods (CUSUM/EWMA) are preferred for low-latency monitoring; segmentation methods (PELT/Binary Segmentation) are preferred for retrospective analysis.

changes may be abrupt, recurrent, or difficult to parameterize [**?** **?**]. When multiple behaviors coexist or regime changes are heterogeneous, even adaptive models can fail. A natural response is therefore to maintain an ensemble of complementary models and adapt at the model level as conditions evolve [**?** **?**].

The core motivation behind ensembles is to provide both robustness and continuity. By maintaining multiple specialized models, the system becomes resilient to unexpected regimes, while preserving knowledge from past conditions that may recur. This property is particularly valuable in finance, where market regimes tend to repeat, and discarding models learned during previous crises can result in the loss of useful information.

As an illustration, consider a trading algorithm with one model optimized for trending markets and another for range-bound markets. An ensemble can monitor a trend indicator and dynamically adjust model usage, either by gradually shifting weights toward the trend model when momentum increases or by switching entirely to the range-bound model during consolidation phases.

In this context, we therefore distinguish three dynamic ensemble mechanisms: (i) gated model selection, (ii) adaptive prediction weighting, and (iii) the incremental introduction and retirement of experts.

Active-model selection (gating). A first strategy is to explicitly choose, at each time step, which model should be active. Several specialized models are maintained (for example, one trained for a stable market and another for a crisis), and a gating mechanism (which may be a regime detector or a learned function) chooses

Table 8: Comparison of adaptation methods for non-stationary financial time series. Methods are categorized by family and compared across key operational dimensions.

Method	Family	Speed	Memory	Complexity	When to Use	Example
Parametric Approaches (Section 5.1)						
TVP / Kalman	Parametric	Fast	Low	$O(p^2)$ per step	Gradual parameter drift, linear models	Dynamic market betas
TAR / STAR	Parametric	Instant	Low	$O(1)$ switch	Known regime driver available	VIX-based volatility regimes
HMM / MS	Parametric	Medium	Medium	$O(KS^2)$	Recurrent discrete regimes	Bull/bear market switching
Dynamic Ensembles (Section 5.2)						
Gating (MoE)	Ensemble	Medium	Medium	$O(M \cdot c)$	Distinct regime specialists	Trend vs range models
Weighted combine	Ensemble	Fast	Medium	$O(M \cdot c)$	Hedge against uncertainty	DWM, Learn++.NSE
Online experts	Ensemble	Slow	High	$O(M \cdot t)$	Evolving regimes, sufficient data	Streaming classifiers
Incremental Learning (Section 5.3)						
SGD + forget	Incremental	Very fast	Very low	$O(p)$ per step	Continuous mild drift	Online linear models
Periodic retrain	Batch	Slow	High	$O(np)$	Stable with periodic shifts	Monthly model refresh
Hyperparameter adapt	Meta	Medium	Low	Varies	Regime-dependent tuning	Learning rate scheduling
Modern Approaches (Section 5.4)						
Continual + EWC	Continual	Medium	Medium	$O(np + p^2)$	Preserve old knowledge	Multi-task fraud detection
Test-time adapt	Online	Fast	Low	$O(k \cdot p)$	Distribution shift, unlabeled test	Domain adaptation
Foundation FT	Transfer	Slow*	High	$O(np_f)$	Limited labeled data	Pre-trained transformers

Notation: p = parameters, S = HMM states, K = iterations, M = ensemble size, c = expert cost, t = training cost, n = samples, p_f = fine-tuned parameters (typically $p_f \ll p$), k = TTA gradient steps.
Speed ratings: Very fast (<1ms), Fast (1–10ms), Medium (10–100ms), Slow (>100ms), Instant (0ms - rule-based).
Memory ratings: Very low ($O(1)$ or $O(p)$), Low ($O(p^2)$ or $O(S)$), Medium ($O(M)$ or $O(np)$), High ($O(nM)$ or full history).
***** Foundation model fine-tuning is slow initially but enables fast subsequent adaptation.

which model to use at each moment [**? ? ? ? ?**]. This is essentially the mixture-of-experts (MoE) idea with a gating network, applied over time [**? ?**]. For instance, a forecasting system may have one expert for low volatility and another for high volatility; a volatility detector (or even the HMM) determines in real time which expert should make the prediction [**? ? ? ?**].

Prediction combination (ensemble weighting). A softer alternative is to avoid hard switching and instead blend the predictions of multiple models. Instead of choosing a single model, combine the outputs of all of them with time-varying adaptive weights. A simple method is to weight models inversely proportional to their recent error — thus, models that are performing poorly (perhaps due to drift) receive lower weight, and if the regime changes and an alternative model starts performing better, the weighting scheme automatically adjusts. This is used in tracking and concept drift meta-learning: algorithms such as Dynamic Weighted Majority (DWM) in the concept-drift literature follow this strategy [**? ? ?**].

Online training of new experts. A more proactive strategy is to expand the ensemble over time by introducing new models as data evolves. In continuous streams, one can continuously train new models on recent windows and add them to the ensemble, possibly removing or "retiring" old models that have become obsolete — a form of continuous learning in which the ensemble grows and is pruned. For example, in the Learn++.NSE [**? ?**] method and variants, for each new batch of data, a new classifier is trained, and the final prediction is a combination of all classifiers with weights that decay for older ones if they misclassify recent data. This gradually "forgets" old hypotheses without discarding them abruptly.

Despite the advantages of ensemble-based approaches, they are not universally appropriate. Their use may be impractical when computational resources are severely constrained, as in ultra–low-latency trading systems, or when strong model interpretability is required for regulatory or operational reasons. Ensembles may also be unnecessary in stable environments where a single well-calibrated model suffices, or undesirable when the operational burden of maintaining and monitoring multiple models becomes prohibitive. In such cases, simpler adaptive approaches—such as incremental learning or time-varying-parameter models—can offer a more suitable balance between adaptability and complexity.

5.3. *Hybrid continuous-adaptation flows*

Hybrid adaptation flows are inherently multi-layered. They combine low-cost, incremental updates operating continuously with explicit responses triggered by detected structural changes. This design seeks to reconcile two competing goals: smooth tracking of minor fluctuations and decisive intervention when regime changes are abrupt or persistent. Figure 16 summarizes a generic hybrid adaptation architecture for learning in non-stationary environments.

The Figure 16 highlights how continuous monitoring, lightweight online updates, and event-driven interventions interact in a single processing flow. The discussion below follows this structure, detailing the main components of such systems and how they jointly balance responsiveness to change with stability over time.

Online monitoring and drift alarms. At the base of the pipeline, change detectors operate continuously, tracking model performance, residuals, or properties of the input data distribution. When these detectors signal a potential structural drift, they act as triggers that activate higher-level adaptation mechanisms.

Local continuous adaptation. Between alarms, the model is kept up to date through lightweight incremental-learning schemes. Typical approaches include exponential forgetting in gradient updates or regular online retraining with newly arriving data. This layer handles small, gradual shifts without destabilizing the model and is supported by many standard learners, such as linear models trained via stochastic gradient descent and neural networks updated through progressive fine-tuning.

Actions upon detecting clear structural changes. When a detector confirms a substantial change, more disruptive actions are initiated, as incremental updates may no longer suffice. Possible responses include partial or total model resets, where prior knowledge is discarded or strongly downweighted; on-the-fly hyperparameter or architecture adjustments tailored to the new regime; or specialization and branching, where a new model is added to an ensemble rather than replacing the existing one. These mechanisms allow flexible responses to heterogeneous or recurring regimes.

Continuity and memory. Adaptation does not imply forgetting. To preserve useful past knowledge, continuity mechanisms from continual learning are often employed. Replay-based strategies, such as maintaining a memory buffer of historical samples and mixing them into retraining, mitigate catastrophic forgetting and enable faster readaptation if previously observed regimes re-emerge.

Supervision and supervised reinitialization. In many high-stakes applications, adaptation is not fully automatic. Drift signals may prompt human review or offline analysis, particularly in regulated domains such as finance. In these cases, experts may recalibrate models, incorporate new explanatory variables, or validate changes under regulatory constraints before redeployment, forming a hybrid manual–automatic loop.

These approaches are constrained by practical tradeoffs. Strong adaptation actions increase computational cost and latency, whereas purely incremental updates may respond too slowly to abrupt changes. The appropriate balance depends on the cost of model error relative to the cost and delay of adaptation, as summarized in Table 9.

Adaptation should not be automatic in all situations. It may be suppressed when drift is transient, when data after change are insufficient, when transaction or operational costs dominate, or when regulatory or stability constraints require fixed models. In such cases, delayed or conservative adaptation can be preferable to rapid but unreliable updates.

In summary, hybrid adaptation architectures provide a principled way to manage non-stationarity by coordinating continuous learning with selective intervention. Their effectiveness lies not only in how they adapt, but also in deciding when adaptation should be limited or deferred, ensuring robustness without unnecessary instability

Table 9: Cost-benefit trade-offs for adaptation strategies in financial applications. Performance gains are approximate and context-dependent; values shown are typical for financial time series forecasting and classification tasks.

Strategy	Latency	Compute Cost	Data Requirements	Perf. Gain	When to Use
No adaptation	0 ms	None	N/A	Baseline (0%)	Stable regimes only; benchmark
Incremental update	1–10 ms	Very low	Streaming only	+5–10%	Gradual drift; HFT; latency-critical
Hyperparameter tuning	1–10 min	Medium	Recent window (100–1000)	+10–15%	Moderate regime change; periodic maintenance
Ensemble reweight	10–100 ms	Low	Recent window (50–500)	+10–20%	Uncertain regimes; multiple hypotheses
Add ensemble member	10 min–1 hr	High	New regime data (500–5000)	+15–25%	Novel regime emerges; sufficient data
Full retrain	1–24 hrs	Very high	Full history (5000+)	+20–30%	Abrupt structural break; offline batch
Model reset	10 min–1 hr	High	New regime only (1000+)	+15–25%	Complete regime change; history irrelevant
Test-time adapt	10–100 ms	Low–medium	Unlabeled test batch	+5–15%	Domain shift; batch inference mode
Foundation fine-tune	1–12 hrs*	Very high	Limited labels (100–1000)	+15–30%	New domain; leverage pre-training

Latency: Time to apply adaptation and resume normal operation. Excludes initial training.
Compute cost: Relative CPU/GPU requirements. "Very low" = single-core CPU sufficient; "Very high" = multi-GPU or distributed cluster.
Data requirements: Approximate number of observations needed for reliable adaptation. Varies by problem dimensionality.
Performance gain: Typical improvement over no-adaptation baseline, measured by accuracy, RMSE reduction, or risk-adjusted returns. Highly problem-dependent.
***** Foundation model fine-tuning is expensive initially but enables rapid subsequent adaptation (<1 hr) to new sub-regimes.
HFT = High-frequency trading (microsecond-level latency requirements).

Decision guidance: In practice, hybrid policies are optimal: incremental updates as default (<10ms overhead), triggered actions (ensemble reweight, hyperparameter tuning) for moderate drift, and scheduled full retraining (nightly/weekly) for major regime changes. The cost of error relative to adaptation cost determines aggressiveness: risk management (high error cost) justifies expensive adaptation; informational forecasts (low error cost) favor cheap incremental approaches.

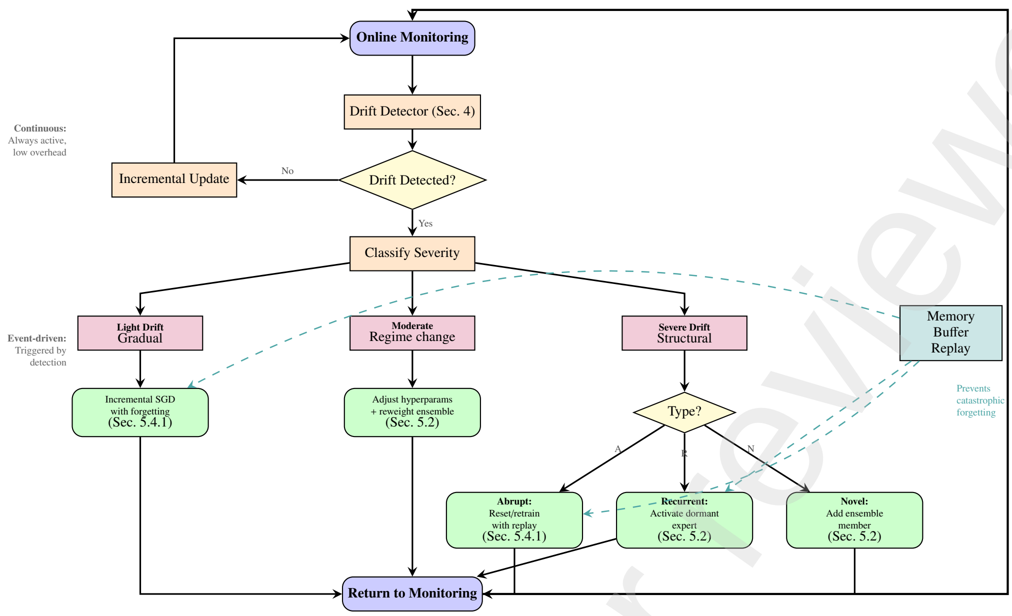

Figure 16: **Hybrid continuous-adaptation pipeline.** The system combines low-overhead incremental updates (continuous monitoring) with event-driven adaptation actions triggered by drift detection. Severity classification routes to appropriate strategies: light drifts receive incremental adjustments, moderate drifts trigger hyperparameter tuning and ensemble reweighting, severe drifts invoke more drastic measures (reset, expert activation, or ensemble expansion). Memory buffers enable replay to mitigate catastrophic forgetting (Sec. 5.4.1). This multi-layer architecture balances adaptation speed, stability, and performance (Sec. 5.3).

5.4. Continuous learning, test-time adaptation, and foundation models

This subsection is organized around three learning strategies for non-stationary environments. We first present continuous learning methods (5.4.1), followed by test-time adaptation techniques (5.4.2). Then, we then discuss light adaptation of foundation models and meta-learning approaches (5.4.3), which aim to enable rapid specialization under distribution shift.

5.4.1. Continuous learning and memory preservation

In continual learning, the objective is to enable models to incorporate new information sequentially while avoiding catastrophic forgetting of previously acquired knowledge [**?**]. This property is particularly important under distribution shift, where adaptation to a new regime should not eliminate the model's ability to operate if earlier regimes reappear or continue to coexist in part of the data.

To achieve this, continual-learning methods rely on mechanisms that constrain how new knowledge is absorbed. Common strategies include parameter regularization, which penalizes changes to parameters that were important for past tasks; replay-based approaches[**?** **?**], which store or generate samples from previous data distributions and mix them into new training; and dynamic architectures, which allocate additional capacity—such as new neurons or modules—to represent emerging regimes while preserving existing ones. In financial settings, for instance, a fraud-detection system that learns a new fraud pattern should not become blind to older, still-relevant fraud behaviors.

A representative example of this philosophy is Elastic Weight Consolidation (EWC) [**?**]. EWC augments the training loss with a penalty that discourages deviations in parameters deemed critical for previous tasks, as measured by Fisher information. In a drift context, once a regime change is identified, the model can be updated using recent data while preserving parameters that encode information from the earlier regime. This allows the model, at least partially, to function across multiple regimes without full retraining.

Replay-based memory further supports this process, especially when historical data are limited or expensive to collect. Maintaining even a small buffer of past observations can substantially reduce forgetting when adapting to new conditions. In practice, this often connects continual learning with ensemble methods: rather than forcing a single model to remember everything, multiple specialized models can be maintained, with the ensemble as a whole preserving broader historical knowledge.

This idea is explicit in data-stream algorithms such as Learn++.NSE and its variants, which continually introduce new models while retaining older ones and adjusting their weights based on current performance. Older hypotheses are not discarded outright, allowing them to remain effective if earlier contexts recur.

Overall, despite these advantages, continual learning has clear limitations. It can underperform when successive regimes are too heterogeneous to be captured by a shared representation, when memory buffers are too small to adequately represent past distributions, or when the model architecture lacks sufficient capacity to express new regime complexity. In addition, if drift occurs faster than the model can adapt—such as during abrupt market shocks—gradual continual-learning updates may lag behind reality. In such cases, ensemble specialization or explicit regime-switching approaches may be more effective than attempting to preserve a single continuously adapting model.

5.4.2. Test-time adaptation

Test-time adaptation (TTA) encompasses methods that allow a model to adjust parts of its behavior during inference itself, using only unlabeled data from the current environment [?]. Rather than accumulating new labeled data and retraining offline, the model performs limited self-adjustment on the fly, aiming to remain aligned with the prevailing data distribution.

At a basic level, TTA often operates by updating internal statistics rather than model parameters. A common example is adaptive normalization: neural networks with normalization layers (e.g., BatchNorm [?] or LayerNorm [?]) rely on estimates of input mean and variance, which may become misaligned under distribution shift. Recomputing or gradually updating these statistics at test time, without modifying the network weights, can already improve robustness to covariate shift.

More advanced TTA methods extend this idea by allowing constrained parameter updates guided by self-supervised objectives defined on test data [?]. Typical choices include entropy minimization or self-consistency criteria, which can be evaluated without labels. The model performs a small number of gradient steps to minimize such objectives before producing predictions. These techniques have shown promise in correcting moderate domain shifts in vision and language models.

In time-series and financial settings, TTA can be interpreted as rapid internal recalibration. For instance, a return-forecasting model trained under one market regime may, when deployed in a new regime, observe incoming inputs and adjust selected parameters or statistics to preserve internal consistency or invariants. While this can provide fast correction without external retraining, it is inherently risky: incorrect self-supervised signals or poorly constrained updates may introduce bias or amplify model drift.

A particularly practical use case arises in anomaly detection, where test data are often assumed to be mostly normal. In such scenarios, thresholds, activity levels, or internal reference statistics can be recalibrated using only current observations. In finance, this is analogous to daily recalibration of risk models using recent intraday prices to maintain alignment with current volatility levels.

Overall, the effectiveness of TTA approaches is limited when shifts are severe or conceptual in nature: if the relationship between inputs and targets changes abruptly, unlabeled test data provide little reliable guidance for adjustment, unless robust self-supervised objectives correlate with the task loss or reliable pseudo-labels can be generated.

5.4.3. Light adaptation of foundation models and meta-learning

Foundation models and meta-learning address non-stationarity by reducing the need for frequent full retraining. Instead of rebuilding models whenever the data distribution changes, they rely on pre-trained representations that can be quickly re-specialized with limited data and computation. This shift is significant in domains such as finance, where full retraining is expensive, slow to deploy, and difficult to govern in production.

This approach is effective under drift for two main reasons. First, pre-training on diverse datasets produces representations that often remain useful when market conditions change, even if the target distribution shifts. Second, adaptation can be limited to a small subset of parameters—such as a task head or parameter-efficient modules like LoRA or adapters—allowing fast updates with bounded computational cost and reduced risk of overfitting or instability [? ? ? ? ?]. As a result, models can respond to regime changes without continuously modifying the full parameter space.

Operationally, a simple deployment loop follows three steps. Incoming data are first encoded into embeddings using a frozen pre-trained backbone. These embeddings are then monitored over time to detect distributional shifts by comparing recent representations against a reference window that characterizes the baseline regime. When a drift detector signals a significant change, a lightweight adaptation step is triggered using a small buffer of recent data. Only the selected parameters—typically the task head or parameter-efficient components—are updated, while the backbone remains fixed. Adaptation is usually constrained by explicit budgets on data, optimization steps, and validation criteria, ensuring that updates improve short-horizon performance and can be safely rolled back if necessary.

This paradigm is already reflected in recent time-series foundation models such as TimesFM, Chronos, and Moirai, which report strong zero-shot and transfer performance across heterogeneous benchmarks. In financial and macroeconomic forecasting, pre-trained forecasters like TimeGPT-1, as well as compact transferable models such as Lag-LLaMA and Tiny Time Mixers, provide practical starting points under common production constraints on latency and compute. In these settings, light adaptation through heads or adapters is often sufficient to maintain performance across changing market conditions, avoiding repeated full retraining cycles.

In summary, foundation models and meta-learning support a controlled and scalable response to non-stationarity. By combining reusable pre-trained representations, continuous monitoring of embedding behavior, and parameter-efficient adaptation, forecasting systems can handle drift more quickly and with lower operational risk. While these methods do not eliminate the need for full retraining in cases of large or persistent shifts, they offer a practical middle ground between static models and costly retraining pipelines, making them well-suited for deployment in evolving financial environments.

6. Evaluation: Protocols and Metrics

This section addresses the research question: *How can we evaluate the performance of models and detection/adaptation systems under non-stationarity, using appropriate metrics and protocols?* Proper evaluation is essential, as inappropriate protocols or metrics can lead to misleading conclusions about the effectiveness of methods in evolving environments.

The discussion is organized around three evaluation components that together define a concise framework for fair and meaningful evaluation under non-stationarity. We first present temporal validation protocols that respect data chronology and avoid information leakage (6.1). We then review metrics specific to change and drift detectors, including detection accuracy, delay, and false alarms (6.2). Finally, we discuss adaptive-performance and cost-aware metrics that capture recovery, stability, and adaptation costs (6.3).

6.1. Temporal validation protocols

In time-series problems, proper evaluation requires validation protocols that respect temporal order, ensuring that no future information is used during training or calibration [**? ?**]. While this principle is conceptually simple, it demands careful implementation in practice: training, validation, and test splits must be defined by contiguous time blocks rather than random partitions, as commonly done under i.i.d. assumptions [**? ?**]. Moreover, when evaluating adaptive models, validation should replicate production conditions; for instance, if a model is updated monthly in deployment, the backtest should explicitly simulate this update cycle [**? ? ?**].

A widely used temporal validation protocol is the *prequential (sequential predictive test)* scheme [**? ? ?**]. In this setting, the model is trained up to a given time and then used to generate predictions sequentially as new observations arrive, with the training set being updated accordingly. This protocol closely mirrors online operation and allows evaluation under realistic data arrival and adaptation conditions [**? ?**].

Beyond general forecasting evaluation, specialized protocols are required for systems that include change detection and adaptation components. For change detection, when real or approximate annotations of changepoint times are available, evaluation is commonly performed by running detectors on series with known changes and observing their behavior relative to these events [**? ?**]. In real-world scenarios where ground truth is unavailable, detectors are often assessed indirectly through their interaction with downstream adaptation mechanisms, for example by embedding them in a full adaptive pipeline and evaluating the resulting system behavior [**? ? ?**].

Similarly, adaptation methods should be evaluated using protocols that emphasize performance across time and regimes rather than relying on static train–test splits [**? ?**]. Common practices include organizing evaluation by temporal windows or regimes and analyzing system behavior before, during, and after distribution shifts, ensuring that adaptation is assessed under conditions that reflect its intended operational use [**?**].

In summary, robust evaluation under non-stationarity relies on temporally consistent validation protocols, realistic simulation of deployment conditions, and end-to-end assessment of detection and adaptation mechanisms within evolving data streams [**? ? ?**].

6.2. *Metrics specific to detectors*

The evaluation of change detectors depends fundamentally on the availability and nature of change-point annotations, which in turn are closely tied to the type of data used. In practice, detectors are evaluated on a spectrum ranging from synthetic and semi-synthetic series to fully real-world data, each offering different levels of control and realism [**? ?**].

A common intermediate setting relies on semi-synthetic datasets, where artificial changes are injected into real time series. This approach provides approximate ground truth while preserving realistic noise characteristics and temporal dependencies, allowing controlled evaluation under conditions that resemble real data [**? ?**]. Fully synthetic series, on the other hand, offer complete control over change locations and mechanisms, but may oversimplify real-world dynamics [**?**].

In contrast, for real-world series without exact changepoint labels, evaluation becomes more challenging. One pragmatic strategy is indirect assessment within an adaptive modeling pipeline: different detectors are combined with the same adaptation policy, and their effectiveness is inferred from downstream behavior [**? ? ?**]. If detector A consistently leads to better-adapted model performance than detector B under identical conditions, it is reasonable to conclude that A is more effective at identifying relevant changes. Finally, in fully unlabeled real data, qualitative inspection or expert judgment is sometimes used, comparing detected changepoints against known historical events—such as the 2008–09 financial crisis or the March 2020 COVID-related market crash—to assess plausibility [**? ?**]. While subjective, this approach can provide contextual validation.

So, when change-point annotations are available, detectors can be evaluated in a supervised manner, similarly to classifiers, by contrasting correct detections with false alarms [**? ?**]. In this setting, several complementary criteria are commonly used to characterize detector behavior [**? ?**].

- **Precision, recall, and F1 for change detection:** these metrics quantify how many true changes are detected (recall) and, among all detected changes, how many correspond to actual changepoints (precision) [**? ?**]. They are particularly useful in scenarios with imbalanced outcomes, where either false alarms or missed detections dominate. In the drift literature, these quantities are sometimes referred to as "drift detection rate" and "false alarm rate" [**? ?**].

- **Mean Time To Detection (MTTD):** beyond whether a change is detected, this criterion captures how quickly the detector reacts. It measures the average delay between the true changepoint and the alarm time [**? ?**]. What constitutes an acceptable delay is application-dependent—for instance, seconds may matter in high-frequency trading, whereas delays of months may be acceptable in macroeconomic analysis.

- **ARL_0 (Average Run Length to false alarm):** this measure focuses on detector behavior under stable conditions. It corresponds to the expected time until a false alarm occurs and is inversely related to the sequential false-positive rate. In practice, detectors are often calibrated by specifying a target ARL_0 [**? ?**].

- **MDR (Missed Detection Rate):** defined as the proportion of true changes that are not detected, this criterion complements recall by explicitly quantifying failures to signal changepoints [**? ?**].

6.3. *Adaptive-performance and cost metrics*

Before analyzing adaptation costs and benefits through specific metrics, it is necessary to define how adaptive models are evaluated in practice. Metrics are only meaningful when computed under evaluation setups that reflect how models are trained, updated, and deployed in a real-world environment over time [**? ? ?**].

Accordingly, the evaluation design must be made explicit. The recommendations below specify how scenarios and protocols may be defined to determine *where* and *under which conditions* adaptive models are tested [**? ?**].

- **Testing across multiple scenarios and datasets:** adaptation should be evaluated on diverse series or assets, covering different temporal structures and drift types, rather than on a single benchmark [**? ? ?**]. Reliance on a small number of canonical benchmarks (e.g., the Electricity dataset[**? ?**] or a few market indices[7]), especially when combined with simplified protocols, may limit the extent to which conclusions generalize [**? ?**].

[7] Examples of widely used market-index data sources include the S&P 500 level series distributed via FRED [**?**]; volatility benchmarks such as the VIX from Cboe (also mirrored in FRED) [**? ?**]; research-grade equity and index returns via CRSP through WRDS [**?**]; and provider methodology/governance documents for major global benchmarks (e.g., MSCI and FTSE Russell) [**? ?**].

- **Robust evaluation protocols:** all methods should be compared under identical backtesting and update procedures, with statistical testing applied where appropriate [?].

- **Sensitivity analysis:** critical parameters controlling adaptation (e.g., forgetting factors, detection thresholds, update frequencies) should be varied to verify that results are not driven by narrow tuning [?].

- **External validity and replicability:** consistent evaluation pipelines and shared datasets enable verification that reported results persist across studies [?].

Once these evaluation setups are fixed, the criteria focus on *what* should be measured to quantify the practical consequences of adaptation. In addition to standard predictive metrics computed sequentially over time—for example, forecasting error tracked across windows or regimes—adaptive models must be assessed with respect to the computational, behavioral, and economic effects introduced by model updates, as summarized below [? ? ?].

- **Computational overhead and latency:** the frequency and duration of model updates determine whether adaptation is feasible under time constraints. Streaming frameworks (e.g., MOA, River) typically report throughput to characterize this aspect [? ?].

- **Memory footprint:** adaptive architectures differ in resource usage; large ensembles require substantially more memory than single-model approaches, which may be prohibitive in constrained environments.

- **Stability and volatility of predictions:** aggressive adaptation can induce erratic behavior. Evaluation should verify that model updates do not introduce excessive oscillations in predictions or decisions over time [? ?].

- **Direct economic impact:** in financial applications, adaptation should be evaluated through risk-adjusted utility measures. When multiple strategies are compared, statistical controls such as White's Reality Check [?] are required to mitigate data-snooping effects [? ?].

7. Benchmark and Reproducibility

This section addresses the research question: *How can detection and adaptation methods be benchmarked under non-stationarity in financial time series?* Proper benchmarking is essential, as inappropriate protocols, scenario choices, or reporting conventions can lead to misleading conclusions about what works in evolving environments [?].

The discussion is organized around four components. We first define benchmark criteria and a scenario specification that makes assumptions explicit and comparable across studies (Section 7.1). We then review datasets and tasks for regime and anomaly detection from both the financial and data-stream literature (Section 7.2). Next, we propose a minimal scenario-coverage baseline that enables comparable benchmark suites without requiring a single canonical dataset (Section 7.3). Finally, we provide a compact recipe for constructing benchmarks and reporting reproducible evaluations under realistic deployment constraints (Section 7.4).

7.1. Criteria and scenario specification

Benchmarks for non-stationarity should support controlled, reproducible comparison across methods and research traditions. In finance, this is challenging because studies often rely on proprietary datasets, ad hoc asset selections, and heterogeneous experimental protocols, which limits reproducibility and undermines cross-paper comparability.

To make benchmark design explicit and comparable, we specify each benchmark scenario s by a *taxonomy-conditioned* descriptor:

$$\phi(s) = (\text{Temporal, Statistical, Spatial, Ontological}), \tag{1}$$

using the four axes introduced in Section 2, together with a data instantiation D_s, an online protocol Π_s, and an evaluation mapping $\mathcal{E}_s$ (Fig. 17):

$$\mathcal{B} = \{(D_s, \phi(s), \Pi_s, \mathcal{E}_s)\}_{s=1}^{S}. \tag{2}$$

Here, Π_s fixes the online setting (e.g., prequential vs. rolling updates, label delay/availability, latency and compute budgets, and model-access assumptions), while $\mathcal{E}_s$ defines what is reported (detection quality, predictive performance and calibration, computational cost, and finance-specific utility).

Under this specification, desirable benchmark properties become requirements on (i) *coverage* of drift signatures $\{\phi(s)\}$ and (ii) *comparability* of protocols and outcomes:

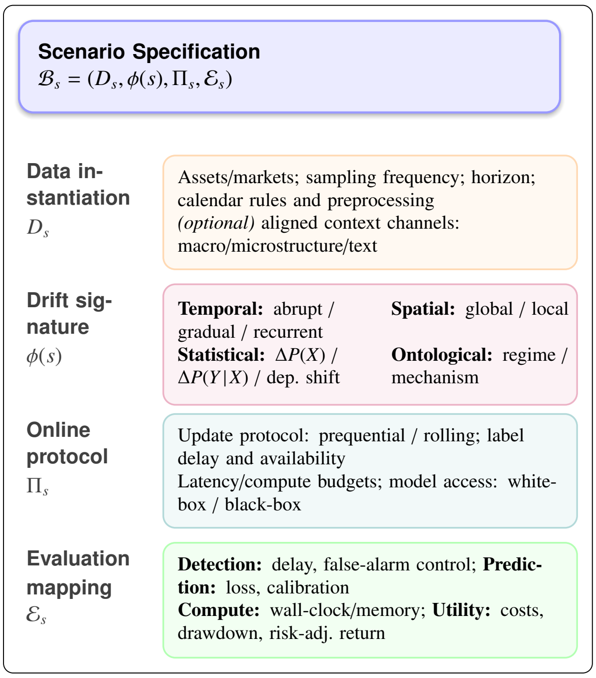

Figure 17: Scenario specification schema for benchmark design under non-stationarity. A scenario is defined by data D_s, drift signature $\phi(s)$ (four-axis taxonomy), online protocol Π_s, and evaluation mapping $\mathcal{E}_s$.

- **Identifiability.** Scenarios should include annotated or controlled changes (event-based, statistical, or semi-synthetic) to enable objective assessment.
- **Coverage.** Suites should span diverse morphologies and mechanisms across the four axes (e.g., abrupt vs. gradual, global vs. local, regime vs. mechanism).
- **Realism–control balance.** Combine real-market episodes with controlled synthetic/semi-synthetic settings to separate methodological effects from idiosyncratic artifacts.
- **Sufficient duration.** Scenarios should be long enough to evaluate long-run stability, repeated updates, and cumulative adaptation effects.
- **Context metadata.** When relevant, provide aligned macro/microstructure/textual context to evaluate context-aware and multimodal approaches [**? ? ?**].

7.2. *Datasets and tasks for regime and anomaly detection*

We organize this discussion into two categories. We first discuss financial time series benchmarks for forecasting and regime change. We then cover anomaly and out-of-distribution benchmarks often used as proxies for rare or extreme regimes. We close with practical considerations on tooling and reproducible pipelines.

7.2.1. *Financial time series and forecasting benchmarks under changing regimes*

In finance, there are still no consolidated benchmark repositories comparable to those used in the data-stream literature. Instead, evaluation typically relies on isolated time series and task-specific experimental setups chosen by individual studies. As a consequence, there is no single "standard dataset" for problems such as regime change or adaptive forecasting: different works select different assets, markets, and time periods. For example, one study may focus on Bitcoin prices [**?**], another on individual stocks [**? ? ? ?**], and another on sector indices, making direct comparison difficult, since results can vary substantially across assets and historical windows.

Several efforts attempt to partially mitigate this limitation by defining approximate regimes using known economic or market events. Common strategies include splitting series into pre- and post-crisis windows (e.g., the 2008 global financial crisis or the 2020 COVID shock) and treating them as distinct regimes to assess robustness under distribution shifts. While useful, these constructions remain ad hoc and lack standardization across studies.

In this fragmented scenario, other types of datasets and tasks are more commonly used, such as:

- Return or volatility forecasting for equity indices (S&P 500, Dow Jones, and international indices) over long horizons, where economic cycles and crises provide implicit regime variation.
- Detection of regime changes in macroeconomic series (GDP, inflation) or market indicators such as implied volatility and trading volume, including interest-rate series affected by monetary policy shifts.
- High-frequency financial data (tick or transaction-level series) used to detect micro-regimes, such as intraday changes in liquidity and volatility; recent examples include microstructure data from B3 [**? ? ?**].
- Specialized tasks such as financial contagion detection, based on time series of correlations or tail-risk measures across markets [**? ? ? ? ?**].

7.2.2. *Benchmarks for anomalies and out-of-distribution*

The distinction between concept drift, regimes, and anomalies is often blurred: rare regimes may appear anomalous when viewed from dominant market conditions, while persistent anomalies can effectively define short-lived regimes. As a result, benchmarks originally designed for anomaly or out-of-distribution (OOD) detection are frequently reused to evaluate drift and regime-detection methods, and the methodological boundaries between these tasks are not always clear.

Several general-purpose benchmarks illustrate this overlap. The Numenta Anomaly Benchmark (NAB) [**?**], for example, includes multiple time series—some with financial relevance, such as stock-related Twitter activity—with annotated anomalies. Although its primary focus is on point anomalies, extended anomalous segments can be interpreted as short regimes, and some studies evaluate drift detectors on NAB by treating each annotated anomaly as a change point to detect. Similarly, recently compiled shift benchmarks such as the Shifts Dataset [**?**] provide real distribution shifts across multiple modalities, even though their emphasis lies mainly outside finance.

In financial applications, however, there is no standardized anomaly or OOD benchmark. Most studies rely on ad hoc, event-based constructions, such as comparing returns during "normal" periods with those observed around major shocks (e.g., September 11 or the Lehman Brothers collapse). While intuitive, these practices lack consistency and comparability across works. As noted by Žliobaitė [**?**] in her critique of the long-standing use of the Electricity data set for concept drift, the widespread adoption of a convenient but limited benchmark can obscure important methodological weaknesses. By analogy, commonly used financial anomaly series—especially those with artificially injected outliers—should be critically examined before being used to evaluate regime-change or drift-detection methods.

7.2.3. *Practical considerations: tooling and reproducible pipelines*

To complement the dataset-and-task survey above, we note that reproducible benchmarking pipelines are often built on: (i) data-stream frameworks (MOA, Scikit-Multiflow, River) for incremental learning, drift detectors, and prequential evaluation [**? ? ?**]; (ii) forecasting toolkits and archives (Monash, GluonTS, Darts) for standardized baselines and dataset access [**? ? ?**]; and (iii) finance-oriented backtesting/simulation environments (Gym-like setups) to evaluate adaptation with economic feedback, including trading frictions and statistical controls against backtest overfitting [**? ? ?**].

These tools facilitate implementation, but fair comparisons require explicit scenario and protocol specifications, which we formalize in Sections 7.3 and 7.4 [**?**].

7.3. *Scenario coverage: towards comparable benchmark suites*

Section 7.2 highlights that, in finance, evaluation is often driven by idiosyncratic choices of assets, time periods, and protocols, which makes results difficult to compare across studies. To address this limitation without requiring a single "standard dataset", we propose to standardize *coverage*: a benchmark suite should include a small set of scenarios whose drift signatures span the main regions of the four-axis taxonomy (temporal, statistical, spatial, ontological), while keeping the suite compact enough for reproducible evaluation.

Minimal reference suite. We recommend the following minimal suite as a *coverage baseline*. Each scenario is defined by a characteristic drift signature $\phi(s)$ (Section 2) and can be instantiated using multiple datasets/tasks from Section 7.2:

- **S1: Crisis dependence regime.** *Abrupt* changes in *multivariate dependence/tails* with *global* impact, interpreted as a *regime* shift (e.g., contagion; correlation/tail-risk breakdown across markets).
- **S2: Local mechanism shift.** *Gradual* change primarily affecting $\Delta P(Y \mid X)$ (often coupled with $\Delta P(X)$), *local* to a subset of assets/segments, interpreted as a *mechanism* change (e.g., sector rotation; microstructure/regulatory change).
- **S3: Secular macro drift.** *Incremental* drift dominated by $\Delta P(X)$ and slow parameter drift with *global* scope, interpreted as *regime drift* (e.g., long-horizon evolution in rates/inflation/volatility regimes).
- **S4: Recurrent seasonal regimes.** *Recurrent* patterns (calendar/intraday seasonality) primarily expressed through $\Delta P(X)$ and higher-order structure; *global* or *local*; typically a *regime* phenomenon.
- **S5: True concept drift (signal efficacy).** *Abrupt or gradual* shifts in $\Delta P(Y \mid X)$ that change the usefulness or sign of predictive relationships; *local or global*; interpreted as *mechanism* or *regime* depending on domain assumptions.

How this complements existing datasets. This suite is intentionally *data-agnostic*: it does not prescribe a single dataset, but specifies what a benchmark suite should cover. For example, index forecasting and macro/indicator monitoring naturally instantiate S3/S4; contagion and correlation-network tasks instantiate S1; microstructure datasets support S2/S4; and strategy/feature instability across regimes targets S5. Anomaly/OOD benchmarks can be used as *proxies* for specific cases (most often S4–S5), but should not be treated as full substitutes for S1–S3 unless they preserve multivariate dependence structure and realistic temporal context.

Minimum reporting for coverage studies. For any suite instantiation $\mathcal{B} = \{(D_s, \phi(s), \Pi_s, \mathcal{E}_s)\}_{s=1}^{S}$, authors should report the scenario specification from Section 7.1 explicitly. At minimum, this includes (i) the intended drift signature(s) $\phi(s)$ (temporal, statistical, spatial, ontological), (ii) the online protocol Π_s (update mode and cadence, label availability/delay, latency/compute budgets, model-access assumptions), and (iii) the evaluation mapping $\mathcal{E}_s$ (detection delay and false-alarm control, predictive loss and calibration, computational cost, and finance-specific utility). To support attribution of improvements, results should include controlled comparisons or ablations that separate representation, detection, and adaptation choices (cf. Section 6).

7.4. Recipe: constructing finance benchmarks under non-stationarity

To enable consistent benchmark construction across datasets and research communities, we propose the following compact recipe. The goal is not to enforce a single dataset, but to enforce *comparable experimental design*.

1. **Choose the target drift signature.** Select a scenario from the coverage suite (Section 7.3) or define a new one by specifying $\phi(s)$ along the four axes (Section 2). This determines what constitutes a "change" and what should be detectable/adaptable.
2. **Instantiate data D_s at the appropriate scale.** Choose markets/assets, sampling frequency, prediction horizon, and any context channels (macro, microstructure, text). Make preprocessing rules explicit (calendar, missing data, corporate actions, normalization).
3. **Specify the online protocol Π_s.** Fix the evaluation mode (prequential vs. rolling), update cadence (time-based or event-based), label availability/delay, and constraints (latency/compute budget; model access such as white-box vs. black-box). Avoid tuning Π_s post hoc to favor a method.
4. **Define the evaluation mapping $\mathcal{E}_s$.** Report at minimum: (i) detection quality (delay and false-alarm control), (ii) predictive performance and calibration under drift, (iii) computational cost (wall-clock/memory, including retraining), and (iv) finance-specific utility (e.g., transaction costs, turnover, drawdown, risk-adjusted return). Use consistent protocols and metrics across methods (Section 6).
5. **Establish baselines and tuning budgets.** Include non-adaptive baselines and simple adaptive baselines (e.g., rolling retrain) to contextualize gains. Specify hyperparameter tuning budgets and calibration targets (e.g., fixed false-alarm rate/ARL0 for detectors) to prevent unfair comparisons.
6. **Reproducibility checklist (minimal).** Provide enough detail to reproduce end-to-end results: data provenance and time span; asset identifiers; exact feature computation windows (and leakage checks); protocol definition (warm-up length, update cadence); detector calibration and thresholds; random seeds; compute environment; and code to run the full pipeline from data to metrics.

8. Discussion and Future Directions

This Section complements the research question: *What are the limitations and future research directions?* The discussion is organized around four classes of threats to validity in learning under non-stationarity. We first examine data- and label-related issues that affect construct validity (8.1). We then analyze evaluation-protocol choices that introduce internal and statistical-conclusion validity threats (8.2). Next, we discuss modeling and adaptation assumptions that can bias conclusions about robustness under drift (8.3). Finally, we address finance-specific limitations—such as the "factor zoo," replicability, and limited generalization—that challenge external validity (8.4).

8.1. Data- and label-related threats (construct validity)

A first family of threats concerns the quality of evaluation data and the definition of "change labels". In real-world financial time series, there is rarely a precise ground truth for when drifts occur. As a result, researchers often rely on proxies, such as associating changes with known events (e.g., market crashes) or injecting synthetic drifts into real data. While pragmatic, these strategies may fail to capture the true nature of non-stationarity, which can directly affect construct validity: the experimental setup may not accurately measure the phenomenon it is intended to assess.

This limitation becomes evident when coarse labels are used to summarize complex dynamics. For example, assigning the entire financial crisis period to a single regime X" and the post-crisis to regime Y" ignores the presence of multiple microdrifts within each phase. In such cases, methods that merely react to large, well-defined shocks may appear overly effective. Similarly, validation on synthetic data with simple linear or abrupt drifts can lead to overly optimistic conclusions, as detectors tuned to these idealized settings often struggle when faced with the nonlinear, overlapping, and gradual changes observed in real markets.

Closely related to this issue is the frequent absence—or low reliability—of validation labels. This is particularly problematic in unsupervised detection settings, where evaluation is often indirect or qualitative. Without objective criteria, performance claims may rely on visual inspection or on the proximity of detected changes to well-known events, leaving room for subjective interpretation and making systematic comparison across methods difficult.

Addressing these challenges opens several directions for future research. One avenue is the development of better-annotated benchmark data sets, possibly combining expert knowledge with data-driven labeling and uncertainty quantification. More generally, evaluation protocols should constrain adaptation rules, computational budgets, and retraining strategies to ensure comparability. Detection frequency, false-alarm cost, and adaptation latency should be reported explicitly. In financial settings, downstream metrics may include risk-adjusted returns, drawdowns, or turnover. This framing shifts evaluation from proxy-based drift detection to robustness and decision-relevant utility under realistic non-stationary conditions.

8.2. Evaluation-protocol threats (internal and statistical-conclusion validity)

A second family of threats stems from inadequate evaluation protocols, which can bias conclusions about drift detectors and adaptation methods. These threats fall into two categories: internal validity, related to experimental control, and statistical-conclusion validity, related to the strength of the empirical evidence.

Internal-validity problems arise when methods are not compared under the same conditions. Common issues include unequal access to data or computation, uneven hyperparameter tuning, and inconsistent temporal protocols. In drift studies, it is frequent that one method is carefully tuned or evaluated without temporal leakage, while others are not. In such cases, observed performance gaps reflect experimental bias rather than methodological superiority.

Statistical-conclusion validity is compromised when performance differences are reported without significance analysis. Many studies claim improvements based on small error reductions or detection gains, without testing whether these differences are robust. In sequential settings, proper tests—such as Diebold–Mariano or paired tests over repeated runs—are required to separate systematic gains from noise. Without them, non-replicable results may be mistaken for progress.

Overall, internal and statistical-validity failures imply that reported gains may disappear under stricter controls or alternative samples, inflating the perceived effectiveness of current methods. In financial applications, these problems can become more serious. Multiple models or variants are often tested on the same historical sample, and only the best result is reported. Apparent improvements are likely driven by chance and overfitting, without correction for multiple testing, for example via White's Reality Check [**?**],

Future research should prioritize controlled and standardized evaluation protocols, in which all methods should share the same data access, tuning budget, and temporal constraints. Performance comparisons must include significance testing and repeated runs. In finance, multiple-testing corrections should be mandatory. Such practices would reduce spurious results and enable more reliable assessment of progress under non-stationarity.

8.3. Modeling- and adaptation-related threats

Another family of threats arises from modeling assumptions and from the design of adaptation mechanisms. To make analysis tractable, many studies adopt simplifying assumptions that rarely hold in operational settings, which weakens both construct and external validity.

A central limitation concerns the assumption of independence within regimes. Much of the concept-drift literature models data as i.i.d. until an abrupt change, after which a new i.i.d. regime is assumed. In contrast, financial time series exhibit strong temporal dependence, including autocorrelation and heteroskedasticity, even within stable regimes.

Beyond temporal dependence, many approaches also assume a fixed feature space. Most methods focus on distribution shifts in the target or in observed covariates, while ignoring feature evolution. In real systems, however, variables may appear, disappear, or change relevance over time.

As a consequence, detectors that ignore temporal structure may respond to volatility clustering or transient dynamics rather than to genuine regime changes, producing misleading signals. When this occurs, a detector may correctly identify a distribution shift but fail to adapt because the informative features themselves are not updated.

These modeling assumptions also propagate directly to adaptation mechanisms. Evaluation setups often presume that retraining or model updates can be performed at negligible cost and without operational side effects. In practice, adaptation introduces latency, computational overhead, and potential instability, especially when labels are delayed or noisy. When such costs are ignored, explains why methods that perform well under controlled benchmarks may fail when exposed to real financial data.

Future research should therefore move toward more realistic modeling and evaluation. Methods should be tested on temporally dependent data with overlapping sources of change, such as concurrent shifts in mean and volatility. Feature evolution should be explicitly modeled rather than assumed away. Finally, adaptation costs, delays, and stability effects should be incorporated into evaluation protocols. Addressing these points is essential for bridging the gap between laboratory performance and real-world robustness.

8.4. *Finance-specific limitations: "factor zoo", replicability, and generalization (external validity)*

In the financial domain, several challenges threaten external validity, that is, the ability of results to generalize across markets and time periods. A key issue is the *factor zoo*, in which many reported factors or strategies show positive historical performance but fail to replicate out of sample. An analogous risk arises for learning-based methods, which may become overfitted to the characteristics of specific markets and time windows rather than capturing broadly stable relationships.

This risk becomes more pronounced in studies that rely on narrow evaluation settings, where a method may perform well on a specific market or historical period but fail to generalize to other assets or time spans. For example, a regime detector could capture bull–bear alternations in US equities between 2000 and 2020 with high accuracy, yet its effectiveness might not carry over to emerging markets or to earlier historical intervals. As financial structures evolve over time, methods that are tightly calibrated to past patterns often struggle to maintain their predictive performance under new conditions.

Future research should prioritize broader and more systematic evaluations across multiple assets and time spans, with explicit attention to replicability. Claims about specific regimes should be tested in analogous settings, and greater emphasis should be placed on structurally grounded approaches that reduce overfitting and enhance generalization, thereby increasing the likelihood that methods maintain their effectiveness outside of the original test conditions.

By confronting these limitations honestly and building robust evidence, the field can progress toward providing finance practitioners with reliable tools to navigate a world of ever-changing data.

9. Conclusion

This research organized the literature on machine learning, econometrics, and quantitative finance into a coherent framework specifically focused on financial time series. We proposed a four-axis taxonomy to characterize non-stationarity, encompassing temporal, statistical, spatial, and ontological dimensions. External representations of non-stationarity were organized around embeddings, multiscale features, and both endogenous and exogenous context. In addition, we structured the literature as an end-to-end pipeline covering drift detection, continuous adaptation, evaluation, and benchmarking.

The study shows that the literature adequately defines the various forms of drift, provides approaches to represent financial series and integrate internal and external information, proposes methods to automatically detect changes, and explores adaptive mechanisms to maintain predictive performance over time. Nevertheless, evaluation and benchmarking practices remain underdeveloped, with limited standardization, weak replicability, and insufficient consideration of operational constraints.

A key insight is that there is no universally optimal detector or adaptation strategy. Method selection must align with the expected type, scale, and dynamics of drift, while deployment decisions must account for computational and operational constraints. Evaluation practices should extend beyond predictive accuracy to include detection delay, false-alarm control, economic utility, and realistic backtesting that incorporates transaction costs and market frictions.

Addressing these gaps requires future research focused on standardized, multi-market benchmarks and decision-centric evaluation protocols that incorporate economic outcomes, implementation costs, and real-world deployment considerations. Taken together, these points highlight that methodological performance cannot be separated from practical feasibility.

In conclusion, by offering a unified taxonomy and structured pipeline, this survey provides a framework to support more comparable evidence, facilitate reliable implementation, and strengthen the feedback loop between methodological advances and practical applications in finance, emphasizing that addressing non-stationarity in financial time series is inherently a systems problem spanning representation, detection, adaptation, and evaluation under practical and economic constraints.

CRediT authorship contribution statement

Davi M. Cabral: Data Curation, Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization.

Adriano L. I. Oliveira: Supervision, Validation, Writing – review.

Gustavo H. F. M. Oliveira: Conceptualization, Methodology, Writing – review & editing.

Adriano Lima: Visualization, Writing – review & editing.

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Data availability

No new datasets were created in this survey. The bibliographic metadata of the reviewed corpus (BibTeX) and machine-readable versions of the survey tables (CSV) are available from the corresponding author upon reasonable request. Code used to generate illustrative plots is publicly available in the authors' repository (see the caption of Fig. 7). Copyrighted full-text articles are not shared. Underlying third-party index data accessed via FRED (e.g., SP500/VIXCLS) are not redistributed.

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